

**2004**

**HIGHER SCHOOL CERTIFICATE  
EXAMINATION**

# Physics

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 21 and 23

**Total marks – 100**

**Section I** Pages 2–26

**75 marks**

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

**Section II** Pages 27–38

**25 marks**

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

## Section I

75 marks

### Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**      $2 + 4 =$      (A) 2     (B) 6     (C) 8     (D) 9  
                                         A ☐     B ☒     C ☐     D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

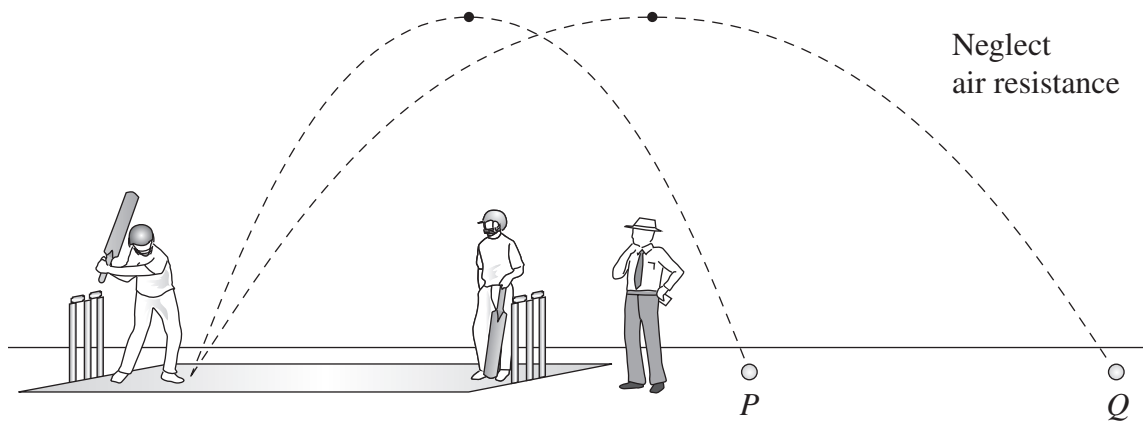
A ☒     B ☒     C ☐     D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒     B ☒     C ☐     D ☐  
                                         *correct*  
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- 1 The picture shows a game of cricket.

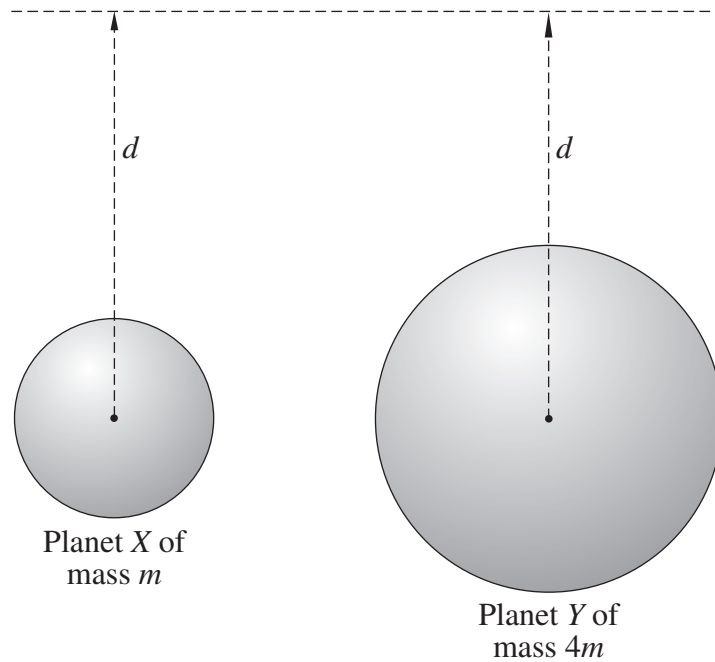


The picture shows two consecutive shots by the batsman. Both balls reach the same maximum height above the ground but ball  $Q$  travels twice as far as ball  $P$ .

Which of the following is DIFFERENT for balls  $P$  and  $Q$ ?

- (A) Time of flight
- (B) Initial velocity
- (C) Gravitational force
- (D) Gravitational acceleration

- 2 The diagram shows two planets  $X$  and  $Y$  of mass  $m$  and  $4m$  respectively.



At the distance  $d$  from the centre of planet  $Y$  the acceleration due to gravity is  $4.0 \text{ m s}^{-2}$ .

What is the acceleration due to gravity at distance  $d$  from the centre of planet  $X$ ?

- (A)  $1.0 \text{ m s}^{-2}$   
(B)  $2.0 \text{ m s}^{-2}$   
(C)  $2.8 \text{ m s}^{-2}$   
(D)  $4.0 \text{ m s}^{-2}$
- 3 A spaceship at a distance  $r$  metres from the centre of a star experiences a gravitational force of  $x$  newtons. The spaceship moves a distance  $\frac{r}{2}$  towards the star.

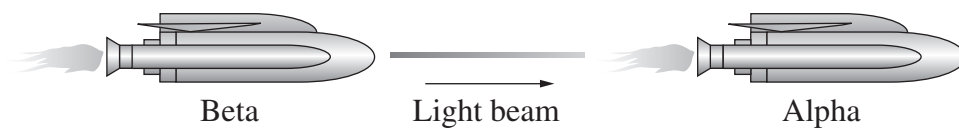
What is the gravitational force acting on the spaceship when it is at this new location?

- (A)  $\frac{x}{2}$  newtons  
(B)  $x$  newtons  
(C)  $2x$  newtons  
(D)  $4x$  newtons

- 4 An object of rest mass  $8.0\text{ kg}$  moves at a speed of  $0.6c$  relative to an observer.

What is the observed mass of the object?

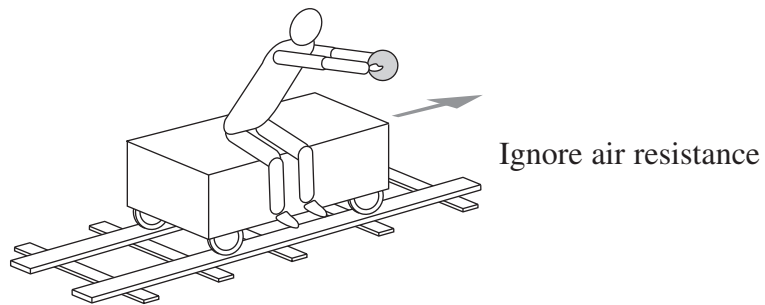
- (A)  $6.4\text{ kg}$
  - (B)  $10.0\text{ kg}$
  - (C)  $12.5\text{ kg}$
  - (D)  $13.4\text{ kg}$
- 5 Two spaceships are both travelling at relativistic speeds. Spaceship Beta shines a light beam forward as shown.



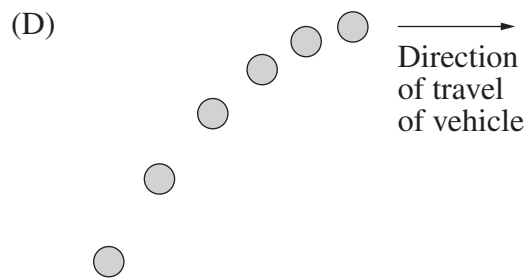
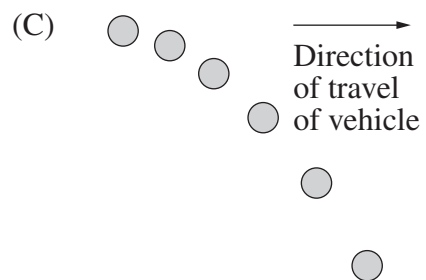
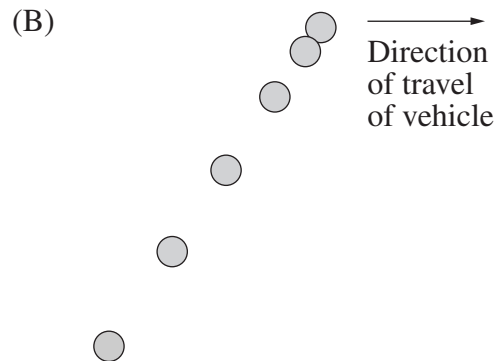
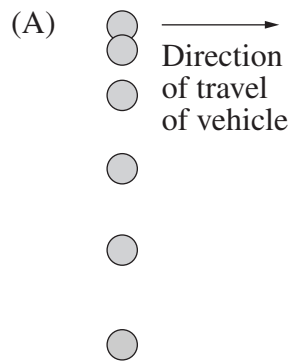
What is the speed of the light beam according to an observer on spaceship Alpha?

- (A) The speed of the light beam is equal to  $c$ .
- (B) The speed of the light beam is less than  $c$ .
- (C) The speed of the light beam is greater than  $c$ .
- (D) More information is required about the relative speed of the spaceships.

- 6 A ball is dropped by a person sitting on a vehicle that is accelerating uniformly to the right, as shown by the arrow.



Which of the following represents the path of the ball, shown at equal time intervals, observed from the frame of reference of the vehicle?



- 7 Why do some electrical appliances in the home need a transformer instead of operating directly from mains power?
- (A) They require a voltage lower than the mains voltage.
  - (B) They require a source of energy that is DC rather than AC.
  - (C) They require an alternating current at a frequency other than 50 Hz.
  - (D) They consume less energy than a similar device without a transformer.
- 8 A transformer which has 60 turns in the primary coil is used to convert an input of 3 V into an output of 12 V.

Which description best fits this transformer?

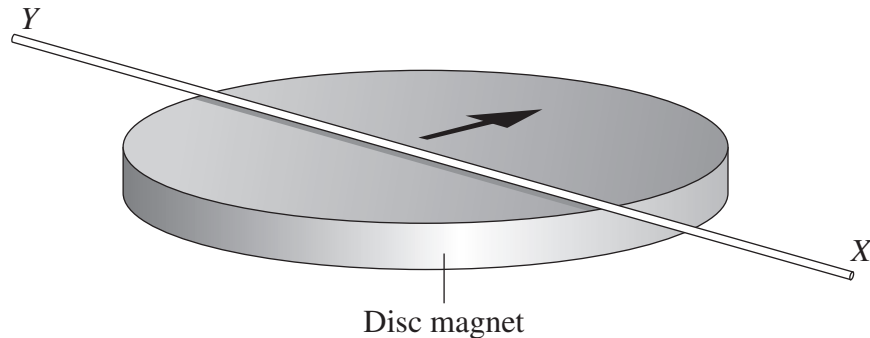
|     | <i>Type of transformer</i> | <i>Number of turns in secondary coil</i> |
|-----|----------------------------|------------------------------------------|
| (A) | Step up                    | 15                                       |
| (B) | Step down                  | 240                                      |
| (C) | Step up                    | 240                                      |
| (D) | Step down                  | 15                                       |

- 9 An electric DC motor consists of 500 turns of wire formed into a rectangular coil of dimensions  $0.2\text{ m} \times 0.1\text{ m}$ . The coil is in a magnetic field of  $1.0 \times 10^{-3}\text{ T}$ . A current of 4.0 A flows through the coil.

What is the magnitude of the maximum torque, and the orientation of the plane of the coil relative to the magnetic field when this occurs?

- (A) 0.04 N m, parallel to the field
- (B) 0.04 N m, perpendicular to the field
- (C) 0.4 N m, parallel to the field
- (D) 0.4 N m, perpendicular to the field

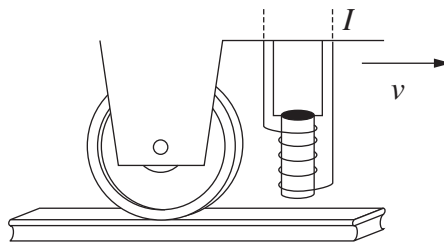
- 10 A disc magnet has its poles on its opposing flat surfaces. An insulated copper wire was placed on the disc magnet as shown in the diagram.



The instant the wire was connected to a DC battery, the wire was observed to move in the direction of the arrow.

Which statement describes the direction of the magnet's field and the direction of the current in the wire, consistent with this observation?

- (A) The field was vertically upward and the current was from  $X$  to  $Y$ .
  - (B) The field was vertically upward and the current was from  $Y$  to  $X$ .
  - (C) The field was in the direction of the arrow and the current was from  $X$  to  $Y$ .
  - (D) The field was in the direction of the arrow and the current was from  $Y$  to  $X$ .
- 11 An electromagnet is attached to the bottom of a light train which is travelling from left to right, as shown.

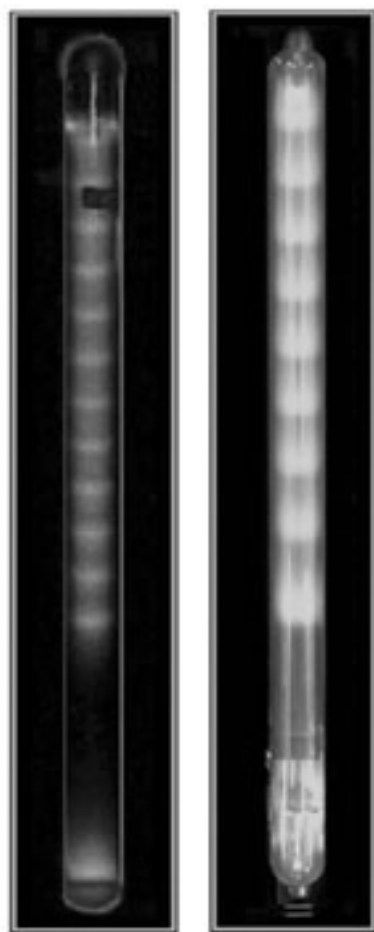


When a large current is passed through the coils of the electromagnet, the train slows down as a direct result of the law of conservation of energy.

In which of the following devices is the law of conservation of energy applied in the same way?

- (A) DC motor
- (B) Loudspeaker
- (C) Induction motor
- (D) Induction cooktop

**12** Photographs of two gas discharge tubes are shown.



What causes the variations of the pattern of striations in the gas discharge tubes?

- (A) Different gases in the tubes
- (B) Different gas pressures in the tubes
- (C) Different voltages applied to the tubes
- (D) Different electrode materials used in the tubes

- 13** Compared to silicon atoms, phosphorus atoms have one more electron in their outer shell. Boron atoms have one less electron than silicon atoms in their outer shell.

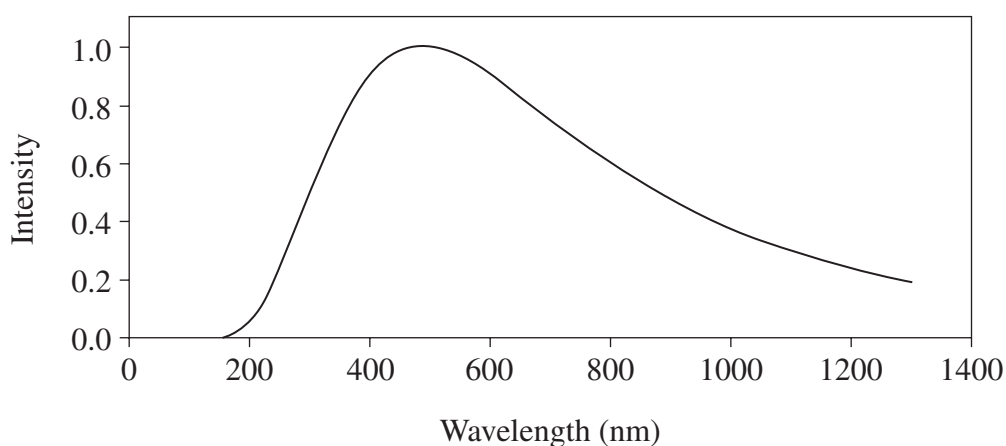
Which of the following is the correct statement?

- (A) An *n*-type semiconductor is produced when silicon is doped with phosphorus, and a *p*-type semiconductor when silicon is doped with boron.
  - (B) A *p*-type semiconductor is produced when silicon is doped with phosphorus, and an *n*-type semiconductor when silicon is doped with boron.
  - (C) The addition of phosphorus atoms turns silicon into a conductor but the addition of boron atoms turns silicon into an insulator.
  - (D) The addition of boron atoms turns silicon into a conductor but the addition of phosphorus atoms turns silicon into an insulator.
- 14** The minimum amount of energy needed to eject an electron from a clean aluminium surface is  $6.72 \times 10^{-19}$  J.

What is the maximum wavelength of incident light that can be shone on this aluminium surface in order to eject electrons?

- (A)  $9.86 \times 10^{-16}$  m
- (B)  $2.96 \times 10^{-7}$  m
- (C)  $3.38 \times 10^6$  m
- (D)  $1.02 \times 10^{15}$  m

- 15 The graph shows the intensity–wavelength relationship of electromagnetic radiation emitted from a black body cavity.



In 1900, Planck proposed a mathematical formula that predicted an intensity–wavelength relationship consistent with the experimental data.

The success of this formula depended on which of the following hypotheses?

- (A) The intensity of light is dependent on the wavelength.
- (B) Light is quantised, with the energy of light quanta depending on the frequency.
- (C) Light is a wave whose intensity is readily expressed using mathematical formulae.
- (D) Light is quantised, with the energy of the light quanta depending on the size of the cavity from which it is emitted.

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## Physics

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Centre Number

## Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

## Question 16 (4 marks)

A projectile is fired at a velocity of  $50 \text{ m s}^{-1}$  at an angle of  $30^\circ$  to the horizontal.

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Determine the range of the projectile.

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**Question 17** (6 marks)

In July 1969 the Apollo 11 Command Module with Michael Collins on board orbited the Moon waiting for the Ascent Module to return from the Moon's surface. The mass of the Command Module was  $9.98 \times 10^3$  kg, its period was 119 minutes, and the radius of its orbit from the Moon's centre was  $1.85 \times 10^6$  metres.

(a) Assuming the Command Module was in circular orbit, calculate

(i) the mass of the Moon; 2

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(ii) the magnitude of the orbital velocity of the Command Module. 2

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(b) The docking of the Ascent Module with the Command Module resulted in an increase in mass of the orbiting spacecraft. The spacecraft remained at the same altitude. 2

This docking procedure made no difference to the orbital speed. Justify this statement.

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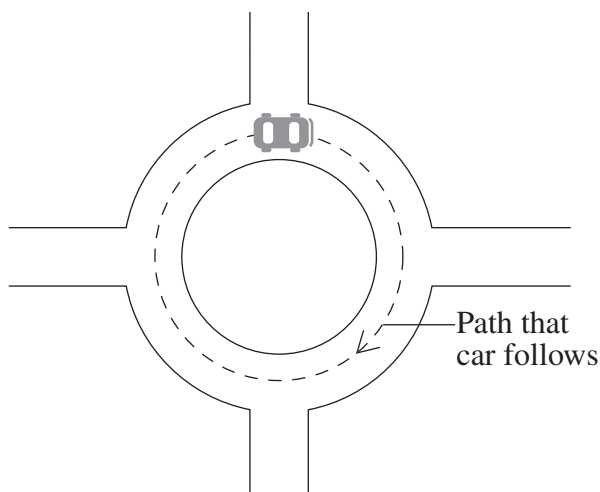
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**Question 18** (4 marks)

A car with a mass of 800 kg travels at a constant speed of  $7.5 \text{ m s}^{-1}$  on a roundabout so that it follows a circular path with a radius of 16 m.

**4**

A person observing this situation makes the following statement.

‘There is no net force acting on the car because the speed is constant and the friction between the tyres and the road balances the centripetal force acting on the car.’

Assess this statement. Support your answer with an analysis of the horizontal forces acting on the car, using the numerical data provided above.

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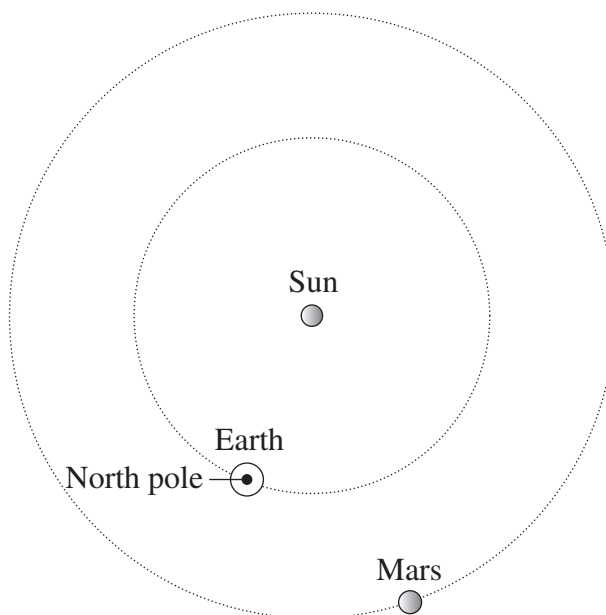
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**Question 19** (6 marks)

On 11 June 2003 the Mars Rover called Spirit was launched on a satellite from Earth when the planets were in the positions shown in the diagram below. The satellite arrived at Mars on 3 December 2003.



- (a) Indicate on the diagram the approximate positions of Earth and Mars on 3 December 2003 and show the satellite's trajectory to Mars. **3**
- (b) Discuss the effect of Earth's motion on the launch and trajectory to Mars of this satellite. **3**

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## Section I – Part B (continued)

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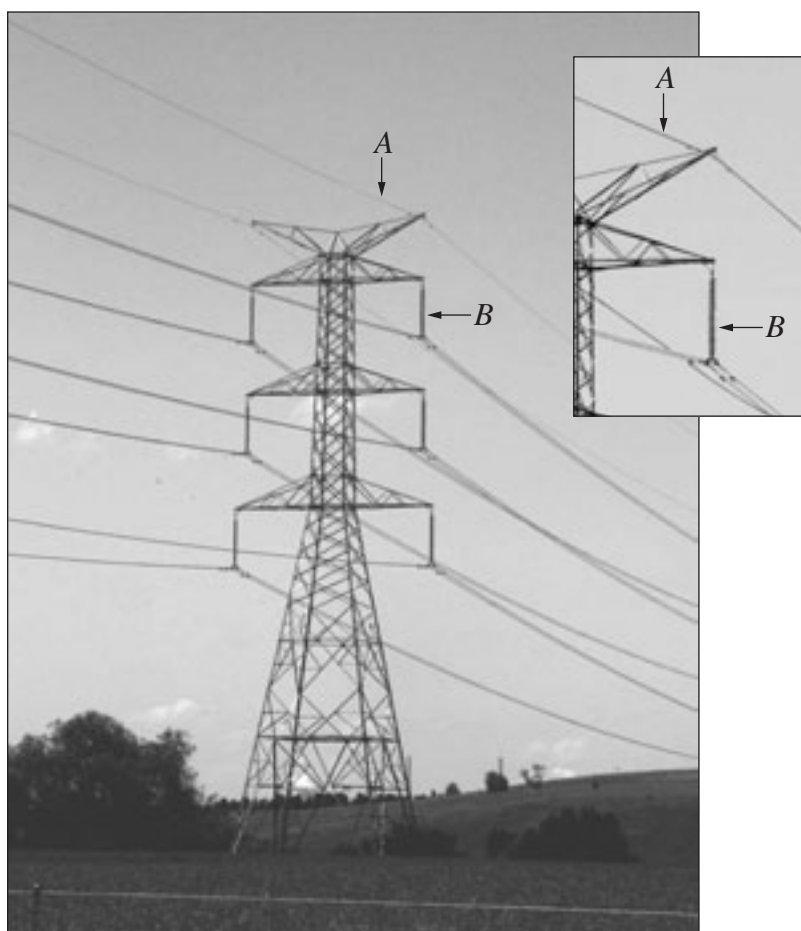
Student Number

Marks

## Question 20 (2 marks)

The photograph below shows a transmission line support tower. The inset shows details of the top section of the tower.

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Describe the role of each of the parts labelled *A* and *B* in the photograph.

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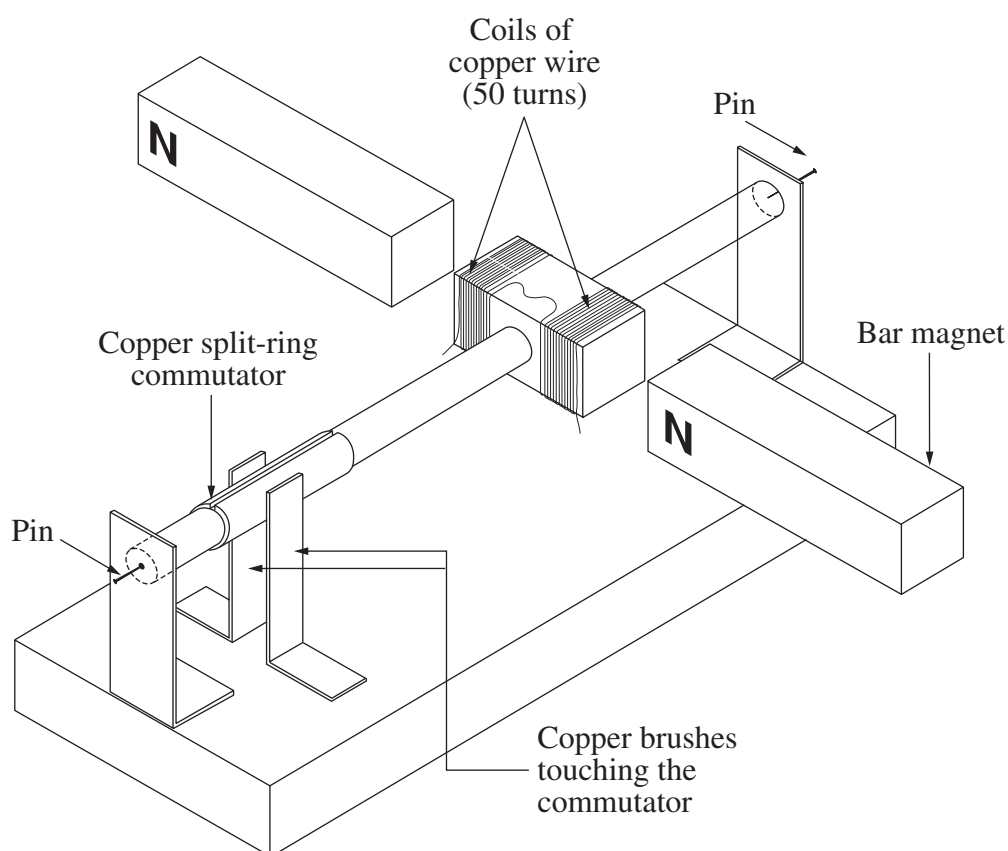
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**Question 21** (6 marks)

(a) The diagram shows a two-pole DC motor as constructed by a student.

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Identify THREE mistakes in the construction of this DC motor as shown in the diagram.

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**Question 21 continues on page 19**

Question 21 (continued)

- (b) An ammeter was used to measure the current through a small DC motor. While it was running freely, a current of 2.09 A was recorded. While the motor was running, the axle of the motor was held firmly, preventing it from rotating, and the current was then recorded as 2.54 A. 3



Explain this observation.

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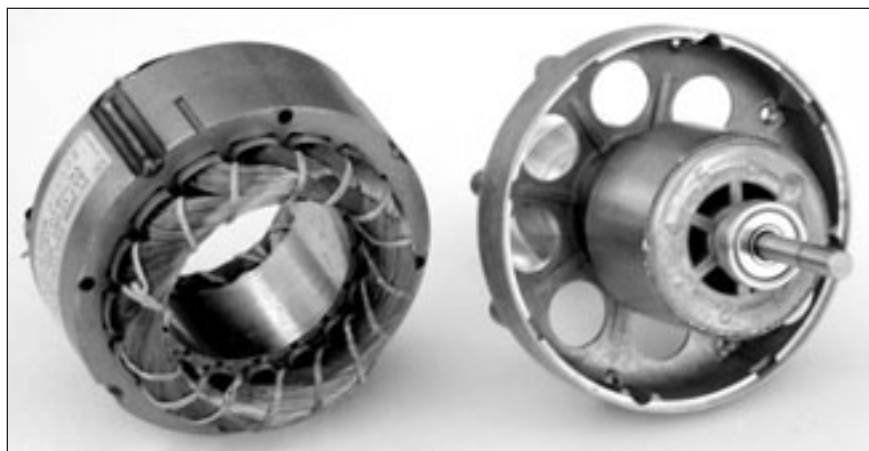
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End of Question 21

**Question 22** (3 marks)

The photograph below shows parts of an AC electric motor.

**3**



Describe the main features of this type of motor and its operation.

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## Section I – Part B (continued)

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**Marks****Question 23** (6 marks)

In the past 50 years electrical technology has developed from the widespread use of thermionic devices to the use of solid state devices and superconductors.

- (a) List THREE disadvantages of thermionic devices that led to their replacement. **3**

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- (b) Outline ONE advantage of using superconductors, with reference to TWO applications. **3**

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6

Propose arguments that Westinghouse could have used to convince authorities of the advantages of his AC system of generation and distribution of electrical energy over Edison's DC supply.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

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## Section I – Part B (continued)

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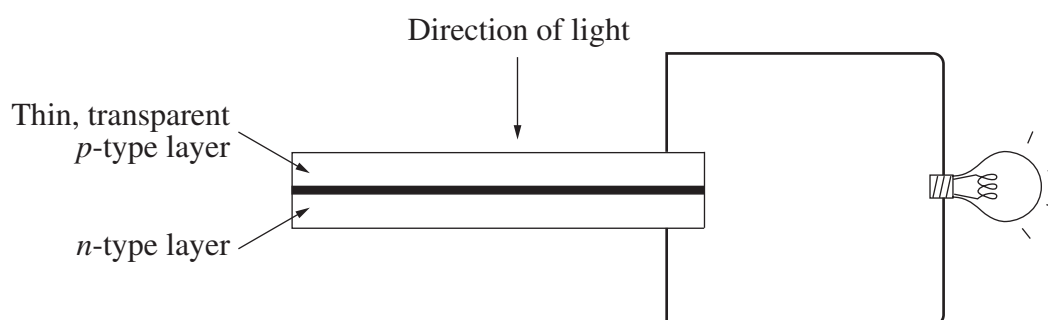
Student Number

Marks

## Question 25 (6 marks)

An example of a solar cell is shown below.

6



The solar cell is able to produce a current due to the photoelectric effect and the electrical properties of the *n*-type and *p*-type layers.

Use this information to outline the process by which light shining on the solar cell produces an electric current that can light up a light globe.

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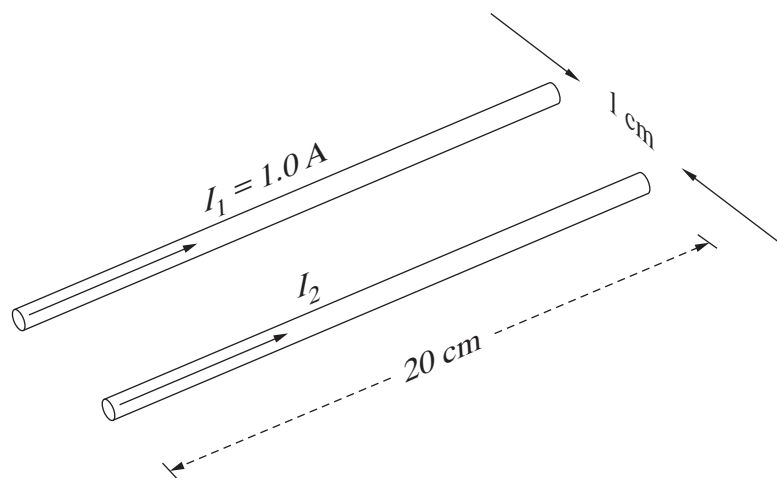
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**Question 26** (7 marks)

The diagram shows part of an experiment designed to measure the force between two parallel current-carrying conductors.



The experimental results are tabulated below.

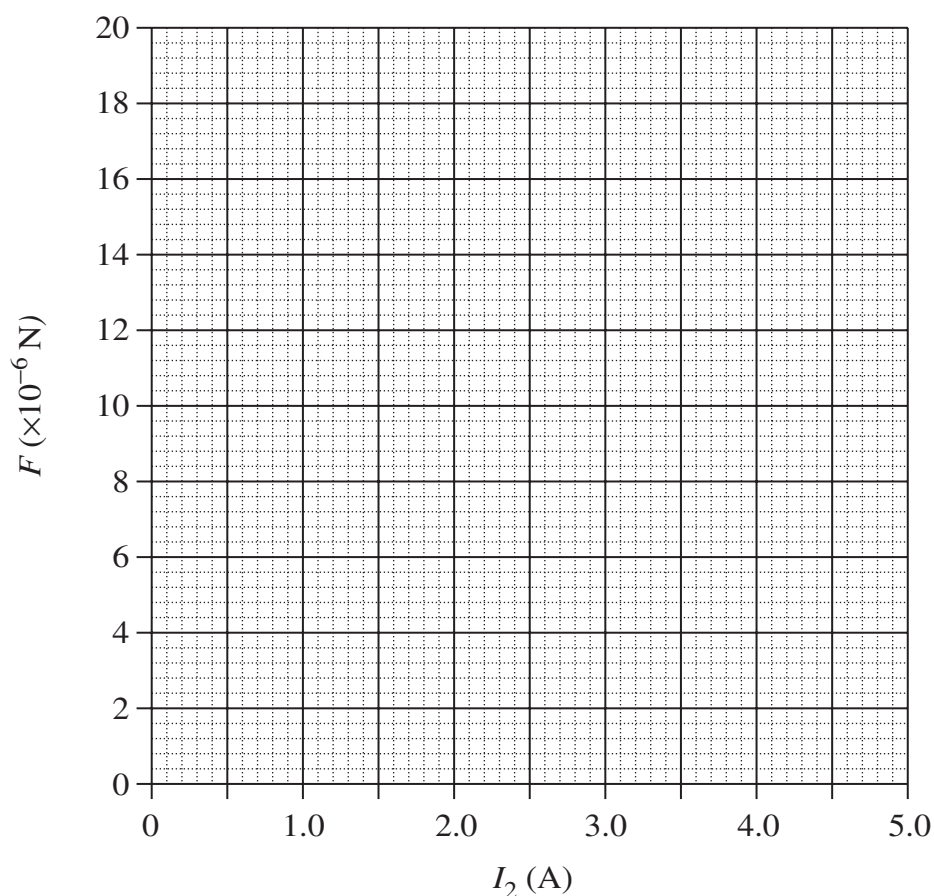
| $I_2 \text{ (A)}$ | Force ( $\times 10^{-6} \text{ N}$ ) |
|-------------------|--------------------------------------|
| 0                 | 0                                    |
| 2.0               | 7                                    |
| 3.0               | 11                                   |
| 4.0               | 14                                   |
| 5.0               | 18                                   |

**Question 26 continues on page 25**

Question 26 (continued)

- (a) Plot the data and draw the line of best fit.

3



- (b) Calculate the gradient of the line of best fit from the graph.

1

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- (c) Write an expression for the magnetic force constant  $k$  in terms of the gradient and other variables.

2

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- (d) Use this expression and the gradient calculated in part (b) to determine the value of the magnetic force constant  $k$ .

1

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End of Question 26

4

Assess the information presented in this magazine, using appropriate calculations to support your argument.

This image shows a full page of handwriting practice paper. It features approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, one above and one below the writing area, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings.

# Physics

## Section II

**25 marks**

**Attempt ONE question from Questions 28–32**

**Allow about 45 minutes for this section**

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

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|                                            | Pages |
|--------------------------------------------|-------|
| Question 28    Geophysics .....            | 28–29 |
| Question 29    Medical Physics .....       | 30–32 |
| Question 30    Astrophysics .....          | 33–34 |
| Question 31    From Quanta to Quarks ..... | 35–36 |
| Question 32    The Age of Silicon .....    | 37–38 |

**Question 28 — Geophysics (25 marks)**

- (a) (i) The magnetic properties of rocks (Earth materials) are useful in the study of geophysics. **1**

Recall TWO other properties of Earth materials that are studied in geophysics.

- (ii) Describe the magnetic properties of Earth materials and outline how these properties have led to an understanding of the variation in Earth's magnetic field over time. **3**

- (b) (i) The period of a simple pendulum can be used to calculate a value for  $g$ , using the relationship **4**

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where  $l$  = length of the pendulum string in metres.

An experiment was performed in which a pendulum 40.0 cm long had a period of 1.268 s.

Use these data to calculate a value for  $g$  and hence calculate the radius of Earth at this location.

- (ii) The pendulum was moved to a new location on the surface of Earth at the same latitude and same distance from the centre of Earth. At this new location the pendulum had a longer period. **2**

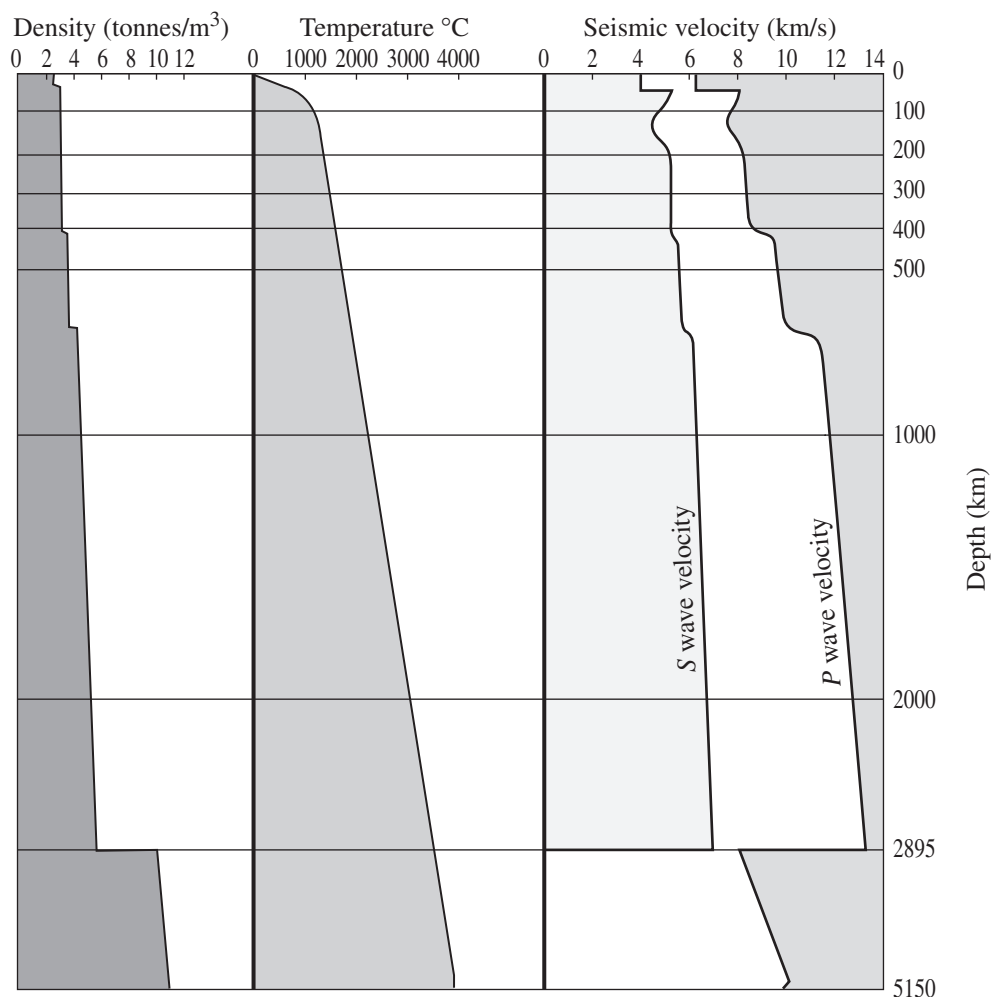
Account for its longer period with reference to Earth's gravitational field and propose a physical basis for this variation.

- (c) Explain the uses of satellites in providing information about Earth. Include in your answer a comparison of geostationary and low Earth orbits, proposing which would be preferred for remote sensing. **7**

**Question 28 continues on page 29**

Question 28 (continued)

(d) The diagram below summarises the changes of properties with depth in Earth.



- (i) During your study of geophysics you carried out a first-hand investigation to analyse the variation in density of different rock types. 3

Describe how your investigation was carried out to ensure that the densities you determined were reliable.

- (ii) Referring to the density graph above, account for the discontinuities (abrupt changes) at 50 km and 2895 km. 2
- (iii) The right-hand section of the diagram shows the velocity of *P* waves and *S* waves. 3

Account for the changes in velocity shown, including an explanation for the effects at 50 km and 2895 km for both *P* and *S* waves.

End of Question 28

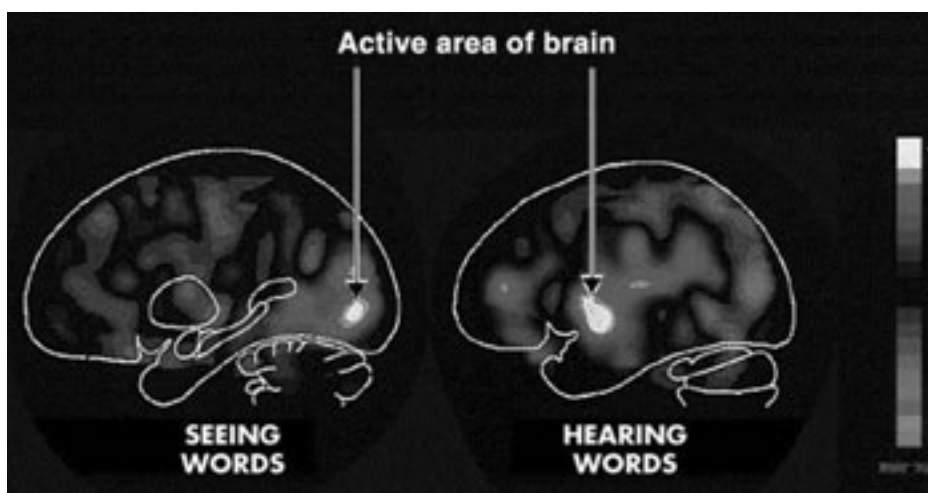
**Question 29 — Medical Physics (25 marks)**

- (a) (i) Describe how the piezoelectric material used in an ultrasound transducer can be made to vibrate to produce compressions and rarefactions in body tissues. **1**
- (ii) Examine the following image showing the heads of unborn twins. **3**



Describe how the image of the heads of the twins was produced.

- (b) Different medical imaging technologies are used to enhance the information available to scientists and doctors. **3**
- (i) The following PET images of the brain show the active areas when the same words were seen on a video screen (left image) and heard through earphones (right image). To produce these images, glucose tagged with the radioisotope F-18 was first injected into the person's body.



With reference to these images and the role of the tagged glucose, evaluate how PET imaging technology is changing our understanding of the way the brain functions.

**Question 29 continues on page 31**

## Question 29 (continued)

- (ii) Identify the imaging technology used to obtain blood flow characteristics of blood moving through the heart, and describe the principle that enables information about the movement of blood to be measured. **3**
- (c) Nobel Prizes are awarded annually ‘to those who . . . have conferred the greatest benefit to mankind’ (quote from Alfred Nobel’s will). The following table shows information about some people who have received Nobel Prizes, and the reasons for their award. **7**

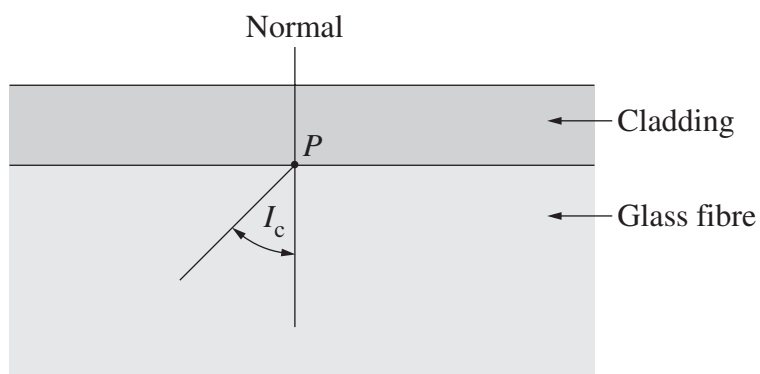
| <i>Award</i>                 | <i>Recipients</i>                                                   | <i>Citation (reasons for award)</i>                                                      |
|------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1956 Nobel Prize for Physics | William Bradford Shockley<br>Walter Houser Brattain<br>John Bardeen | ‘for their researches on semiconductors and their discovery of the transistor effect’    |
| 1972 Nobel Prize for Physics | John Bardeen<br>Leon Neil Cooper<br>John Robert Schrieffer          | ‘for their jointly developed theory of superconductivity, usually called the BSC-theory’ |
| 2003 Nobel Prize for Physics | Alexei Abrikosov<br>Vitaly Ginzburg<br>Anthony Leggett              | ‘for their pioneering contributions to the theory of superconductors and superfluids’    |
| 2003 Prize for Medicine      | Peter Mansfield<br>Paul Lauterbur                                   | ‘for their discoveries concerning magnetic resonance imaging’                            |

With reference to the physical processes upon which MRI depends, assess the impact of advances in knowledge about semiconductors and superconductors on the development of magnetic resonance imaging.

**Question 29 continues on page 32**

## Question 29 (continued)

- (d) (i) During your study of medical physics you carried out a first-hand investigation of the transfer of light by optical fibres. The diagram below shows part of the cross-section of an optical fibre, with the critical angle labelled. 2



Sketch the diagram in your answer booklet and show a ray of light that is totally internally reflected at the point  $P$  in the fibre.

- (ii) The photograph shows a normal endoscopic image of the transverse part of the large intestine. 3



[www.gastrolab.net](http://www.gastrolab.net)

Describe how the optical fibres in an endoscope are used to produce an image such as the one shown.

- (iii) Describe how an endoscope could be used to obtain tissue samples from inside the large intestine, and outline why the endoscope is of particular use in this procedure. 3

**End of Question 29**

**Question 30 — Astrophysics (25 marks)**

- (a) (i) Identify the initial and final elements of the principal sequence of nuclear reactions for a star on the Main Sequence. **2**
- (ii) Identify the type of star that the Sun will initially turn into after it completes its Main Sequence evolution. State the main source of energy in the core at this stage. **2**
- (b) The apparent magnitudes of three stars are measured with a telescope equipped with a camera, first with a red filter placed in front of the detector, and then with a blue filter placed in front of it. The absolute magnitudes of the three stars can be determined from their spectra, and are listed in the fourth column of the table for the red filter.

The results are shown in the table.

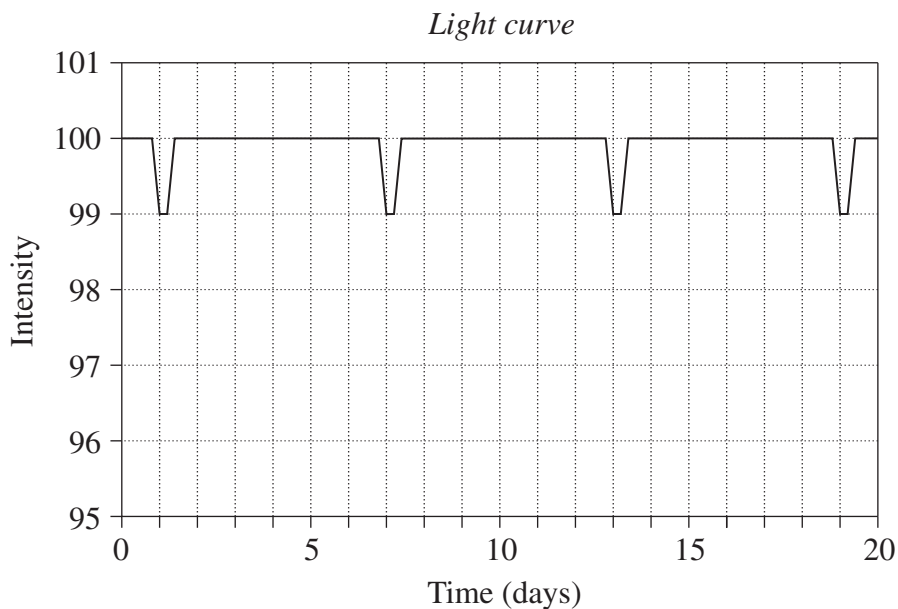
| <i>Star</i> | <i>Apparent magnitude<br/>red filter</i> | <i>Apparent magnitude<br/>blue filter</i> | <i>Absolute magnitude<br/>red filter</i> |
|-------------|------------------------------------------|-------------------------------------------|------------------------------------------|
| Betelgeuse  | −0.89                                    | +0.41                                     | −6.47                                    |
| Rigel       | +0.18                                    | +0.14                                     | −6.69                                    |
| Sirius      | −1.46                                    | −1.46                                     | +1.46                                    |

- (i) Use the data in the table to determine which is the bluest of these three stars. **3**
- (ii) Calculate the distance to Rigel in parsecs. **3**
- (c) Describe how the spectrum of a star can be used to determine its temperature, chemical composition and aspects of its motion. **7**

**Question 30 continues on page 34**

## Question 30 (continued)

- (d) An astronomer made regular measurements of the intensity of a star over the course of several days and obtained the light curve shown below.



- (i) Describe the features of this light curve that suggest that the astronomer is observing an eclipsing binary system. **2**
- (ii) If both stars have equal masses of  $2 \times 10^{30}$  kg, determine the separation of the two stars. **3**
- (iii) The astronomer concludes that the system is a white dwarf eclipsing the other star. The intensity of light from the star is proportional to its cross-sectional area. **3**

That is,  $I \propto \pi r^2$ .

Using the data and diagram, calculate the radius of the white dwarf as a fraction of the radius of the other star. Assume that the white dwarf has negligible luminosity.

**End of Question 30**

## Question 31 — From Quanta to Quarks (25 marks)

- (a) (i) Identify TWO features of the strong nuclear force that binds the nucleons together within the nucleus of an atom. 2
- (ii) When Chadwick discovered the neutron he estimated its mass as 1.15 times the mass of the proton, quite close to its true value. 2

State the TWO laws of physics he used to make this estimate.

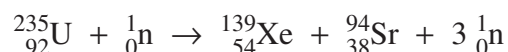
- (b) (i) The table below lists the first generation of quarks and antiquarks. 2

| <i>Quarks</i> |               |                 | <i>Antiquarks</i> |               |                 |
|---------------|---------------|-----------------|-------------------|---------------|-----------------|
| <i>Name</i>   | <i>Symbol</i> | <i>Charge</i>   | <i>Name</i>       | <i>Symbol</i> | <i>Charge</i>   |
| Up            | u             | $+\frac{2}{3}e$ | Antiup            | $\bar{u}$     | $-\frac{2}{3}e$ |
| Down          | d             | $-\frac{1}{3}e$ | Antidown          | $\bar{d}$     | $+\frac{1}{3}e$ |

The Standard Model of matter states that baryons, like protons and neutrons, are comprised of three quarks, while mesons, like the pions  $\pi^+$  and  $\pi^-$ , are comprised of one quark and one antiquark.

Using the table above, state the quark composition of the neutron and the negative pion.

- (ii) The first atomic bomb was a simple uranium-235 fission device. One mode of fission for uranium-235 is given below. 4



Calculate the mass defect and the energy released per  ${}^{235}\text{U}$  atom, given the following nuclear masses and other data:

$${}_{92}^{235}\text{U} = 234.9934 \text{ u}$$

$${}_{54}^{139}\text{Xe} = 138.8883 \text{ u}$$

$${}_{38}^{94}\text{Sr} = 93.8945 \text{ u}$$

$${}_0^1\text{n} = 1.00867 \text{ u}$$

$$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$$

$$c = 3.00 \times 10^8 \text{ ms}^{-1}$$

$$\text{u} = \text{atomic mass unit}$$

Question 31 continues on page 36

Question 31 (continued)

(c)

7

One cannot understand the [particle] physics of the past several decades without understanding the nature of the accelerator . . . the dominant tool in the field for the past forty years. By understanding the accelerator, one also learns much of the physics principles that physicists have laboured centuries to perfect.

Leon Lederman and Dick Teresi, *The God Particle*, 1993

Describe how the key features and components of the standard model of matter have been developed using accelerators as a probe.

- (d) (i) During your study of From Quanta to Quarks you carried out a first-hand investigation to observe the visible components of the hydrogen spectrum. 2

Identify the equipment you used to observe this spectrum.

- (ii) During your physics course you examined first hand the emission spectrum of atomic hydrogen. The four coloured lines are listed in the table below. 4

| <i>Colour of the emission line</i> | <i>Name of the emission line</i> | <i>Electron transition</i> |
|------------------------------------|----------------------------------|----------------------------|
| Red                                | $H_{\alpha}$                     | $n = 3$ to $n = 2$         |
| Green                              | $H_{\beta}$                      | $n = 4$ to $n = 2$         |
| Blue                               | $H_{\gamma}$                     | $n = 5$ to $n = 2$         |
| Violet                             | $H_{\delta}$                     | $n = 6$ to $n = 2$         |

Calculate the wavelength of the  $H_{\beta}$  spectral line, and hence determine the energy of the emitted photon.

- (iii) Describe TWO limitations of Bohr's model of the hydrogen atom. 2

**End of Question 31**

**Question 32 — The Age of Silicon (25 marks)**

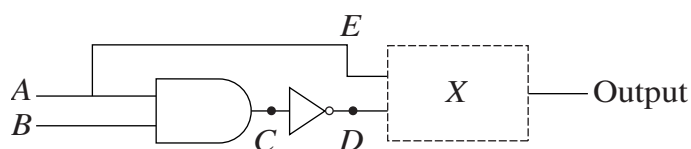
(a) (i) Outline the role of the electromagnet and switch contacts in a relay. **2**

(ii) Explain how a relay works. **2**

(b) (i) Identify the gate shown below and predict the output if the input at *A* is 1 and at *B* is 0. **2**



(ii) The diagram below shows a logic circuit. Determine the gate *X* which gives an output of 1 if the input *A* is 1 and *B* is 1. Justify your answer by using a truth table. **4**



(c) SILIAC, the first computer owned and operated by the University of Sydney, built in the 1950s, was constructed using thermionic devices. Its successor, the KDF9, was built in the late 1960s using solid state devices (transistors). Today's supercomputers are built using integrated circuits. **7**

Assess the impact on computers of each succeeding device, with reference to the differences between each device.

**Question 32 continues on page 38**

## Question 32 (continued)

- (d) The diagrams below show two different inverting amplifiers.

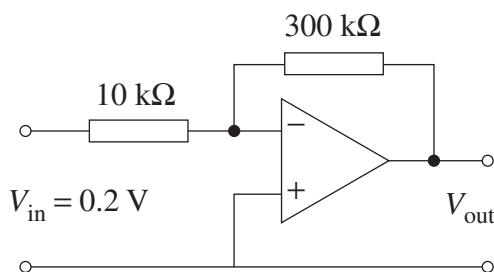


Diagram 1

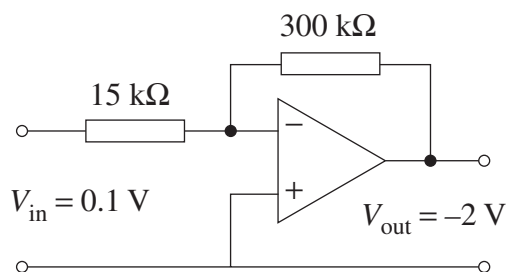


Diagram 2

- (i) Calculate  $V_{\text{out}}$  in Diagram 1.

2

The two circuits are now combined to produce a summing amplifier as shown below.

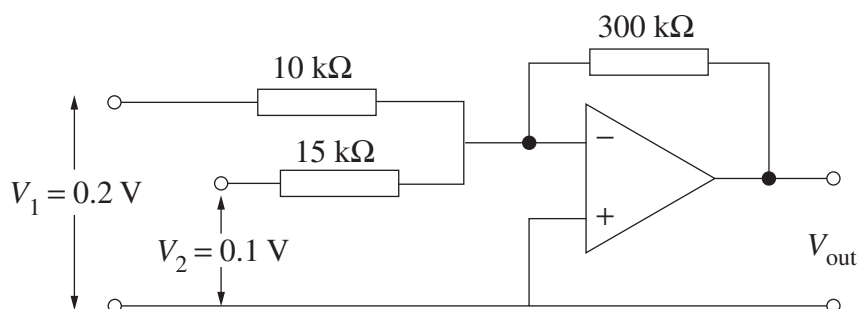


Diagram 3

- (ii) Calculate  $V_{\text{out}}$  in Diagram 3 using your results in part (i) and the data in Diagram 2, and verify your value at  $V_{\text{out}}$  using the following formula for the output voltage for a summing amplifier.

3

$$V_{\text{out}} = -R_3 \left( \frac{V_1}{R_1} + \frac{V_2}{R_2} \right).$$

- (iii) Explain the use of the three resistors in the summing amplifier shown in Diagram 3.

3

**End of paper**

## DATA SHEET

|                                                                     |                                                               |
|---------------------------------------------------------------------|---------------------------------------------------------------|
| Charge on electron, $q_e$                                           | $-1.602 \times 10^{-19} \text{ C}$                            |
| Mass of electron, $m_e$                                             | $9.109 \times 10^{-31} \text{ kg}$                            |
| Mass of neutron, $m_n$                                              | $1.675 \times 10^{-27} \text{ kg}$                            |
| Mass of proton, $m_p$                                               | $1.673 \times 10^{-27} \text{ kg}$                            |
| Speed of sound in air                                               | $340 \text{ m s}^{-1}$                                        |
| Earth's gravitational acceleration, $g$                             | $9.8 \text{ m s}^{-2}$                                        |
| Speed of light, $c$                                                 | $3.00 \times 10^8 \text{ m s}^{-1}$                           |
| Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$ | $2.0 \times 10^{-7} \text{ N A}^{-2}$                         |
| Universal gravitational constant, $G$                               | $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$          |
| Mass of Earth                                                       | $6.0 \times 10^{24} \text{ kg}$                               |
| Planck constant, $h$                                                | $6.626 \times 10^{-34} \text{ J s}$                           |
| Rydberg constant, $R$ (hydrogen)                                    | $1.097 \times 10^7 \text{ m}^{-1}$                            |
| Atomic mass unit, $u$                                               | $1.661 \times 10^{-27} \text{ kg}$<br>$931.5 \text{ MeV}/c^2$ |
| 1 eV                                                                | $1.602 \times 10^{-19} \text{ J}$                             |
| Density of water, $\rho$                                            | $1.00 \times 10^3 \text{ kg m}^{-3}$                          |
| Specific heat capacity of water                                     | $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$           |

## FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \quad \text{therefore} \quad a_{\text{av}} = \frac{v-u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2} a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

## FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left( \frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left( \frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

| PERIODIC TABLE OF THE ELEMENTS |                                |                                |                                |                                |                                 |                                   |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |  |  |  |  |
|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|---------------------------------|-----------------------------------|--------------------------------|------------------------------|--------------------------------|-------------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|--------------------------------|---------------------------------|---------------------------------|------------------------------|--|--|--|--|
| KEY                            |                                |                                |                                |                                |                                 |                                   |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |  |  |  |  |
|                                |                                | Atomic Number                  |                                | 79<br>Au<br>197.0              |                                 | Symbol of element                 |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |  |  |  |  |
|                                |                                | Atomic Weight                  |                                | Gold                           |                                 | Name of element                   |                                |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |  |  |  |  |
| 1<br>H<br>1.008<br>Hydrogen    | 4<br>Be<br>9.012<br>Beryllium  | 5<br>B<br>10.81<br>Boron       | 6<br>C<br>12.01<br>Carbon      | 7<br>N<br>14.01<br>Nitrogen    | 8<br>O<br>16.00<br>Oxygen       | 9<br>F<br>19.00<br>Fluorine       | 10<br>Ne<br>20.18<br>Neon      |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |  |  |  |  |
| 11<br>Na<br>22.99<br>Sodium    | 12<br>Mg<br>24.31<br>Magnesium | 13<br>Al<br>26.98<br>Aluminium | 14<br>Si<br>28.09<br>Silicon   | 15<br>P<br>30.97<br>Phosphorus | 16<br>S<br>32.07<br>Sulfur      | 17<br>Cl<br>35.45<br>Chlorine     | 18<br>Ar<br>39.95<br>Argon     |                              |                                |                               |                              |                              |                                |                               |                                |                                 |                                 |                              |  |  |  |  |
| 19<br>K<br>39.10<br>Potassium  | 20<br>Ca<br>40.08<br>Calcium   | 21<br>Sc<br>44.96<br>Scandium  | 22<br>Ti<br>47.87<br>Titanium  | 23<br>V<br>50.94<br>Vanadium   | 24<br>Cr<br>52.00<br>Chromium   | 25<br>Mn<br>54.94<br>Manganese    | 26<br>Fe<br>55.85<br>Iron      | 27<br>Co<br>58.93<br>Cobalt  | 28<br>Ni<br>58.69<br>Nickel    | 29<br>Cu<br>63.55<br>Copper   | 30<br>Zn<br>65.39<br>Zinc    | 31<br>Ga<br>69.72<br>Gallium | 32<br>Ge<br>72.61<br>Germanium | 33<br>As<br>74.92<br>Arsenic  | 34<br>Se<br>78.96<br>Selenium  | 35<br>Br<br>79.90<br>Bromine    | 36<br>Kr<br>83.80<br>Krypton    |                              |  |  |  |  |
| 37<br>Rb<br>85.47<br>Rubidium  | 38<br>Sr<br>87.62<br>Strontium | 39<br>Y<br>88.91<br>Yttrium    | 40<br>Zr<br>91.22<br>Zirconium | 41<br>Nb<br>92.91<br>Niobium   | 42<br>Mo<br>95.94<br>Molybdenum | 43<br>Tc<br>[98.91]<br>Technetium | 44<br>Ru<br>101.1<br>Ruthenium | 45<br>Rh<br>102.9<br>Rhodium | 46<br>Pd<br>106.4<br>Palladium | 47<br>Ag<br>107.9<br>Silver   | 48<br>Cd<br>112.4<br>Cadmium | 49<br>In<br>114.8<br>Indium  | 50<br>Sn<br>118.7<br>Tin       | 51<br>Sb<br>121.8<br>Antimony | 52<br>Te<br>127.6<br>Tellurium | 53<br>I<br>126.9<br>Iodine      | 54<br>Xe<br>131.3<br>Xenon      |                              |  |  |  |  |
| 55<br>Cs<br>132.9<br>Caesium   | 56<br>Ba<br>137.3<br>Barium    | 57–71<br>Lanthanides           |                                | 72<br>Hf<br>178.5<br>Hafnium   | 73<br>Ta<br>180.9<br>Tantalum   | 74<br>W<br>183.8<br>Tungsten      | 75<br>Re<br>186.2<br>Rhenium   | 76<br>Os<br>190.2<br>Osmium  | 77<br>Ir<br>192.2<br>Iridium   | 78<br>Pt<br>195.1<br>Platinum | 79<br>Au<br>197.0<br>Gold    | 80<br>Hg<br>200.6<br>Mercury | 81<br>Tl<br>204.4<br>Thallium  | 82<br>Pb<br>207.2<br>Lead     | 83<br>Bi<br>209.0<br>Bismuth   | 84<br>Po<br>[210.0]<br>Polonium | 85<br>At<br>[210.0]<br>Astatine | 86<br>Rn<br>[222.0]<br>Radon |  |  |  |  |
| 87<br>Fr<br>[223.0]            | 88<br>Ra<br>[226.0]            | 89–103<br>Actinides            |                                | 104<br>Rf<br>[261.1]           | 105<br>Db<br>[262.1]            | 106<br>Sg<br>[263.1]              | 107<br>Bh<br>[264.1]           | 108<br>Hs<br>[265.1]         | 109<br>Mt<br>[268]             | 110<br>Uun<br>—               | 111<br>Uuu<br>—              | 112<br>Uub<br>—              | 113<br>Uuq<br>—                | 114<br>Uuq<br>—               | 115<br>Uup<br>—                | 116<br>Uuh<br>—                 | 117<br>Uus<br>—                 | 118<br>Uuo<br>—              |  |  |  |  |
|                                |                                |                                |                                |                                |                                 |                                   |                                |                              |                                | Ununbium                      |                              | Ununquadium                  |                                | Ununhexium                    |                                |                                 |                                 | Ununoctium                   |  |  |  |  |

Lanthanides

|                                |                             |                                   |                                |                                   |                               |                               |                                 |                              |                                 |                              |                             |                              |                                |                               |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|-----------------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|
| 57<br>La<br>138.9<br>Lanthanum | 58<br>Ce<br>140.1<br>Cerium | 59<br>Pr<br>140.9<br>Praseodymium | 60<br>Nd<br>144.2<br>Neodymium | 61<br>Pm<br>[146.9]<br>Promethium | 62<br>Sm<br>150.4<br>Samarium | 63<br>Eu<br>152.0<br>Europium | 64<br>Gd<br>157.3<br>Gadolinium | 65<br>Tb<br>158.9<br>Terbium | 66<br>Dy<br>162.5<br>Dysprosium | 67<br>Ho<br>164.9<br>Holmium | 68<br>Er<br>167.3<br>Erbium | 69<br>Tm<br>168.9<br>Thulium | 70<br>Yb<br>173.0<br>Ytterbium | 71<br>Lu<br>175.0<br>Lutetium |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|-----------------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|

Actinides

|                                 |                              |                                   |                             |                                  |                                  |                                  |                               |                                  |                                    |                                    |                                 |                                     |                                  |                                    |
|---------------------------------|------------------------------|-----------------------------------|-----------------------------|----------------------------------|----------------------------------|----------------------------------|-------------------------------|----------------------------------|------------------------------------|------------------------------------|---------------------------------|-------------------------------------|----------------------------------|------------------------------------|
| 89<br>Ac<br>[227.0]<br>Actinium | 90<br>Th<br>232.0<br>Thorium | 91<br>Pa<br>231.0<br>Protactinium | 92<br>U<br>238.0<br>Uranium | 93<br>Np<br>[237.0]<br>Neptunium | 94<br>Pu<br>[239.1]<br>Plutonium | 95<br>Am<br>[241.1]<br>Americium | 96<br>Cm<br>[244.1]<br>Curium | 97<br>Bk<br>[249.1]<br>Berkelium | 98<br>Cf<br>[252.1]<br>Californium | 99<br>Es<br>[252.1]<br>Einsteinium | 100<br>Fm<br>[257.1]<br>Fermium | 101<br>Md<br>[258.1]<br>Mendelevium | 102<br>No<br>[259.1]<br>Nobelium | 103<br>Lr<br>[262.1]<br>Lawrencium |
|---------------------------------|------------------------------|-----------------------------------|-----------------------------|----------------------------------|----------------------------------|----------------------------------|-------------------------------|----------------------------------|------------------------------------|------------------------------------|---------------------------------|-------------------------------------|----------------------------------|------------------------------------|

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.  
The atomic weights of Np and Tc are given for the isotopes <sup>237</sup>Np and <sup>99</sup>Tc.

**2004 HSC Notes from  
the Marking Centre  
Physics**

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# 2004 HSC NOTES FROM THE MARKING CENTRE

## PHYSICS

### Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Physics. It provides comments with regard to responses to the 2004 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

It is essential for this document to be read in conjunction with the relevant syllabus, the 2004 Higher School Certificate Examination, the Marking Guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Physics.

### General Comments

In 2004, 10 158 candidates attempted the Physics examination.

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately.

Overall, the candidates' responses were appropriate and indicated a level of understanding of physics concepts that is appropriate for HSC candidates. Candidates need to be reminded that the answer space allocated is a guide to the maximum length of response required. Similarly, the key word used in the question gives an indication of the depth of the required response. The Option question is divided into a number of parts. Candidates should clearly label each part of the question when writing in their answer booklets.

The space provided should be used as a guide to the length of the response required. Additional (blank) spaces may be used to plan or draft responses. By providing additional information beyond the requirements of the question, candidates may disadvantage themselves by introducing contradictions. Candidates should, where possible, make use of clearly labelled diagrams to answer and/or supplement a concise response.

Show all relevant working in questions involving calculations. An incorrect answer standing alone cannot score marks. Responses showing correct equations and substitutions but incorrect calculations may score some marks.

The use of formulae as supplied on the formulae sheet before substitution and rearrangement may be of benefit to candidates. Similarly candidates are advised to refer to the data sheet. Candidates are also reminded to be aware of the appropriate units used in formulae.

**Section I – Core****Part A**

| Question | Correct Response |
|----------|------------------|
| 1        | B                |
| 2        | A                |
| 3        | D                |
| 4        | B                |
| 5        | A                |
| 6        | B                |
| 7        | A                |
| 8        | C                |

| Question | Correct Response |
|----------|------------------|
| 9        | A                |
| 10       | A                |
| 11       | C                |
| 12       | B                |
| 13       | A                |
| 14       | B                |
| 15       | B                |
|          |                  |

**Part B****Specific Comments****Question 16**

A substantial number of candidates were able to gain full marks for this question.

Most responses correctly resolved the initial velocity vector into its vertical and horizontal components. Better responses used the vector components in the relevant equations and identified the direction of the acceleration.

**Question 17**

The better responses demonstrated a sound understanding of physics and were able to link the content drawn from several dot points, thus combining relevant equations to demonstrate the relationships involved.

- (a) (i) Better responses converted time to seconds, rearranged terms successfully and made no arithmetical or transcription errors. A significant number of responses showed an incorrect selection of equations from the data.
- (ii) Many responses incorrectly used the formula for escape velocity rather than calculating orbital velocity from the equations for centripetal and gravitational force.
- (b) In the better responses, candidates demonstrated their understanding of the link between orbital velocity and altitude. They were then able to use this to support their argument.

**Question 18**

The better responses succinctly identified the two clear errors in the statement and used the centripetal force equation to support their judgement. Many candidates incorrectly interpreted the statement as being correct and therefore had difficulty as they tried to justify its elements.

### Question 19

- (a) The better responses recognised that Mars travelled approximately a third of a revolution in the time that Earth completed half a revolution. Many responses incorrectly showed both planets orbiting clockwise as viewed in the diagram. Poorer responses had trajectories that did not show the Sun at one focus of an elliptical path.
- (b) Candidates are reminded that the gain in velocity due to the slingshot effect as a satellite passes by a planet does not provide velocity boost to the launch of a rocket from the Earth. Many responses clearly identified that both the Earth's rotation on its axis and revolution around the Sun contributed to the increased velocity of the rocket on launch but did not relate these to the effect of the Earth's motion on the trajectory.

### Question 20

Candidates are reminded to identify and use the key word at the beginning of the question to construct an answer. A significant proportion of candidates could not identify the labelled parts.

### Question 21

- (a) Good responses clearly identified the mistakes in the motor's construction. Poorer responses described parts of the motor, but not mistakes in its construction. A significant number of candidates could identify two errors in the design but only a small proportion could identify three.
- (b) A significant number of candidates did not recognise this as a question about back EMF and so offered explanations dealing with Ohm's law and the power equation. The better responses related the back EMF generated when the coil was spinning to the voltage supplied to the motor. Some weaker responses, in explaining the observation using back EMF, equated the back EMF to a current rather than a voltage.

### Question 22

A significant number of candidates recognised the motor in the photographs. In weaker responses, candidates knew the names of the parts of the motor but some had trouble describing their functions. The operation of the motor was, in general, not well described.

In a large number of responses the features of different motors were mixed, adding features from a synchronous motor (slip rings) when describing an induction motor. Candidates are reminded to plan their response to directly address the question and present information logically.

### Question 23

- (a) Better responses showed understanding of the terms *thermionic devices*, *semiconductors* and *superconductors*.
- (b) Better responses presented a scientifically accurate, coherent answer that provided the main features of one advantage with its respective applications. Many responses showed a lack of understanding of the distinction between the terms *advantages* and *applications*. Poorer responses showed limited understanding of the operation of maglev trains.

Candidates are reminded to address the question and present relevant information in an organised response. A significant number of candidates gave a detailed explanation of the BCS theory, which was not required.

### Question 24

Most candidates demonstrated a familiarity with the competition between Westinghouse and Edison and were able to show an understanding of the differences between AC and DC generation and distribution.

A majority of responses proposed arguments that Westinghouse could have used to promote AC over DC with the better responses focused on both generation and distribution. A significant number of responses proposed arguments for distribution of AC but did not deal with generation at all.

Most candidates displayed a sound understanding of the function of the transformer in the process of distribution of AC but few candidates could explain the advantages of the AC generator over the DC generator.

Weaker responses gave a list of features and/or advantages of AC but were unable to link the advantages to the features to form arguments.

### Question 25

Most candidates were able to sketch in general terms the main features of the solar cell. Better responses clearly identified the correct sequence of steps from the initial light being directed onto the solar cell until the light globe lights up.

Better responses related the electrical properties of the n-type and p-type layers to the creation of a potential difference in the solar cell.

Weaker responses confused the direction of electron flow in the circuit with that in the semiconductor layers.

### Question 26

- (a) Well answered by nearly all candidates. A significant number of candidates did not plot the first point (0,0).
- (b) A significant number of responses did not include the correct order of magnitude for the value of force and consequently that of the gradient.
- (c) Many responses identified the appropriate equation. Poorer responses showed a lack of skill in manipulating the equation to obtain the required expression. Some responses contained substitutions into the equation that indicated a lack understanding of the intent of the question.
- (d) In a number of responses, the magnitude for the value of the gradient, although not stated in part (b), was correctly used in part (d).

**Question 27**

Most candidates recognised this as a projectile motion question. A large number of responses used the appropriate equations to carry out calculations but many of these responses contained errors in the substitution of data. In better responses, a valid judgement was made using appropriate calculations to support the arguments. An example of such a response follows.

Now,  $v_y = u_y + at$ . If Jordan stayed in the air for 2.5 seconds, then by 1.25 seconds, he reaches maximum height.  
 $\therefore 0 = u_y + 1.25 \times -9.8, \therefore u_y = 12.25 \text{ m/s}$

However,  $\Delta y = u_y t + \frac{1}{2} a_y t^2$

$$\begin{aligned} \therefore \Delta y &= 12.25 \times 1.25 + \frac{1}{2} \times 9.8 \times 1.25^2 \\ &= 7.66 \text{ m} \end{aligned}$$

This states that Jordan reaches a maximum height of 7.66m, which is impossible for a human with physical legs! Since the basket of the basketball is between 2 and 3 metres it would take less than 1 second overall to reach a height of 3 metres above the ground. Therefore the magazine is exaggerating TOO MUCH and the information stated in the information is not valid at all.

**Section II – Options****Question 28 – Geophysics**

- (a) (i) Well answered by the majority of candidates.
- (ii) Better responses linked magnetic mineral reversals with magnetic field reversals.
- (b) (i) Most responses involved the correct determination of ‘g’ using the supplied equation, although many candidates had difficulty in selecting the appropriate equation to calculate the Earth’s radius.
- (ii) Good responses supplied reasons for the longer period by relating it to changes in the gravitational field due to properties of the underlying materials.
- (c) Better responses contained descriptions of how satellites provide information in remote sensing, and included good comparisons for geostationary versus low earth orbit satellites. Poorer responses did not relate the types of data with types of satellite.
- (d) (i) Good responses demonstrated an understanding of the investigation technique including experimental factors involved and evidence of reliability. Poorer responses gave partial descriptions of possible experiments.
- (ii) Better responses answered the question with clear references to the supplied data.

- (iii) Better responses contained references to the structures of the earth's interior.

### Question 29 – Medical Physics

- (a) (i) The better responses were short and referred to high frequency alternating potential difference being applied across the piezoelectric material. Many candidates confused AC with alternating potential difference and a significant few indicated that ultrasound was produced when current went through the crystal.
- (ii) While most responses recognised that the image was an ultrasound and described the process of its production in general terms, better responses related reflections to differences in acoustic impedances of materials at boundaries between tissues. Many responses referred to differences in density rather than acoustic impedance and most did not relate the production of the image to time differences in the transducer receiving reflected ultrasound nor did they relate the brightness of the image to the amount of reflection.
- (b) (i) Better responses provided clear descriptions of the physical processes and then made judgements based on them.
- (ii) While most responses identified the process as Doppler ultrasound, and could describe the Doppler effect with examples, the better responses related specified changes in frequency or wavelength of ultrasound to the direction of blood flow. In some responses it seemed evident that some candidates thought that b (ii) was about PET.
- (c) Many good responses demonstrated a sound knowledge of the physical process of MRI and linked this development to the development of semiconductors and superconductors.

A significant number of poorer responses either did not contain information about semiconductors or superconductors or did not refer to the MRI process or indicated a lack of understanding of what 'physical process' meant. Many candidates wrote unnecessarily long responses to this question.

- (d) (i) Most of candidates demonstrated an understanding of the angle of incidence being equal to the angle of reflection when they drew a ray of light being totally internally reflected. However, a surprising number of candidates did not draw their incident ray with an incident angle greater than the critical angle. A significant number of candidates did not sketch the diagram faithfully into the answer booklet and subsequently did not convincingly draw the incident ray at an angle greater than the critical angle. It appeared that many candidates mistook the line in the diagram in the question, which represented the critical angle to be the actual incident ray, and they simply drew a reflection of this ray in their sketched answer.
- (ii) This part of the question was generally well answered, with most candidates demonstrating clear understanding of the role of non-coherent fibres and coherent fibres in an endoscope. Better responses also showed a clear understanding of the role of the coherent fibre bundles in maintaining the integrity of the image delivered by the endoscope. Poorer responses simply described the role of total internal reflection in the transport of light within optical fibres.

- (iii) The best responses described the incorporation of surgical tools into the endoscope, described the insertion of the endoscope and the procedure involved (colonoscopy) in taking the sample. These responses also outlined the suitability of endoscopes for the procedure in the question by giving reasons such as the avoidance of invasive surgery and the ability of endoscopes to provide the surgeon with a clear view during the procedure.

### Question 30 – Option Astrophysics

- (a)
  - (i) A number of responses indicated that candidates had overlooked the words ‘Main Sequence’, and therefore their responses continued beyond the required hydrogen and helium. In some responses, the term *elements* seemed to be misunderstood, often resulting in involved descriptions of the total evolution of types of star.
  - (ii) Most responses identified that the Sun will initially change into a (Red) Giant after its hydrogen is exhausted. In some responses, however, the word *initially* was interpreted as referring to the initial stage of being a Red Giant, eliciting a different response.
- (b)
  - (i) The more successful responses included a calculation of colour index for each star and then related this to the blueness of a star. Reference to absolute magnitude was inappropriate to this section.
  - (ii) Most responses included an identification of the appropriate formula and correct substitutions. In weaker responses, there was some confusion as to which apparent magnitude to substitute and/or problems in manipulating the equation, including the log, to determine distance.
- (c) The better responses recognised that the spectrum received from any star consists of both a continuous spectrum (similar to a blackbody) and a set of absorption lines. Candidates are advised to clarify their use of words such as *spectra* in contexts such as this. Some responses excelled by the use of appropriate labelled diagrams or graphs of various kinds, including the H–R diagram.
- (d)
  - (i) Candidates performed well in this part of the question. Good responses related features of the light curve to justify a binary system. In better responses candidates succinctly identified that the light curve had ‘periodic dips in intensity’. In poorer responses a binary system was described rather than the features of the light curve necessary to identify the presence of a binary system.
  - (ii) Candidates are reminded that this part of the question required a change of units for period from days to seconds and then squaring it.
  - (iii) Candidates were challenged by this question, having difficulty using the relationship provided to determine the answer. Poorer responses indicated that there was some confusion in understanding of the terms luminosity, magnitude and intensity

### Question 31 – Quantum to Quarks

- (a) (i) Well answered. Most candidates could name two features of the strong nuclear force.
- (ii) Well answered. Most candidates correctly identified the required laws. Conservation of mass was a common error.
- (b) (i) Well answered. Most responses demonstrated that the correct answers could be deduced from the tabulated data.
- (ii) Most responses indicated an understanding of the need to calculate a difference in the masses before and after the fission occurs. Better responses contained few arithmetic errors. Simple errors in addition and subtraction were common.
- (c) Better responses provided a succinct description that considered both sides of the question, the standard model and particle accelerators, and particularly how they were linked together as asked in the question. Poorer responses contained the history of the atomic model but stopped short of the ‘standard model’. There was also some confusion between neutron scattering and particle acceleration. Many candidates wrote unnecessarily long responses to this question.
- (d) Candidates are reminded that practical experiences are a mandatory part of the course and are examinable.
  - (i) Better responses named the required equipment without describing the experiment.
  - (ii) Candidates tended to answer this part of the question better than other parts. Poorer responses did not contain the second calculation.
  - (iii) A majority of candidates successfully answered this question. Poorer responses named two limitations but did not describe them.

### Question 32 – Age of Silicon

Most candidates attempted all sections.

- (a) (i) Good responses clearly outlined the function of the parts of a relay, including the use of diagrams.
- (ii) Good responses linked the cause and effect to explain how the relay works.
- (b) (i) Well answered.
- (ii) There was a wide variety of correct responses.
- (c) Most responses provided good specific detail. Good responses contained details as to the specific advantages/disadvantages of each generation of devices and good closing summaries that made judgements about their impact on computers.
- (d) Better responses clearly showed in parts (i) and (ii) the sequence of steps required to calculate and verify  $V_{\text{out}}$ .

# Physics

## 2004 HSC Examination Mapping Grid

| Question                    | Marks | Content                                            | Syllabus Outcomes |
|-----------------------------|-------|----------------------------------------------------|-------------------|
| <b>Section I<br/>Part A</b> |       |                                                    |                   |
| 1                           | 1     | 9.2.2 Col 2 dot 1                                  | H9                |
| 2                           | 1     | 9.2.3 Col 2 dot 2 / Col 3 dot 2, 9.2.1 Col 3 dot 3 | H9, H6, H12       |
| 3                           | 1     | 9.2.3 Col 3 dot 2 / Col 2 dot 3                    | H9, H12           |
| 4                           | 1     | 9.2.4 Col 3 dot 5                                  | H6, H12           |
| 5                           | 1     | 9.2.4 Col 2 dot 9                                  | H6                |
| 6                           | 1     | 9.2.4 Col 2 dot 4                                  | H6, H14           |
| 7                           | 1     | 9.3.4 Col 3 dot 6                                  | H7, H13           |
| 8                           | 1     | 9.3.4 Col 3 dot 2                                  | H7, H12           |
| 9                           | 1     | 9.3.1 Col 3 dot 4                                  | H9, H12           |
| 10                          | 1     | 9.3.1 Col 2 dot 4                                  | H9, H12           |
| 11                          | 1     | 9.3.2 Col 2 dot 5 / Col 3 dot 4, 9.3.5 Col 2 dot 1 | H3, H9            |
| 12                          | 1     | 9.4.1 Col 3 dot 1                                  | H10               |
| 13                          | 1     | 9.4.3 Col 2 dot 6, 7                               | H7                |
| 14                          | 1     | 9.4.2 Col 3 dot 4                                  | H10, H12          |
| 15                          | 1     | 9.4.2 Col 2 dot 3                                  | H10, H1           |
| <b>Section I<br/>Part B</b> |       |                                                    |                   |
| 16                          | 4     | 9.2.2 Col 3 dot 1                                  | H6, H12.4         |
| 17 (a) (i)                  | 2     | 9.2.2 Col 3 dot 5                                  | H9, H6            |
| 17 (a) (ii)                 | 2     | 9.2.2 Col 3 dot 4, 5, 9.2.3 Col 3 dot 2            | H9, H6            |
| 17 (b)                      | 2     | 9.2.2 Col 2 dot 10                                 | H9, H6            |
| 18                          | 4     | 9.2.2 Col 2 dot 8                                  | H6, H13, H14      |
| 19 (a)                      | 3     | 9.2.2 Col 2 dot 6, dot 10                          | H6, H13, H14      |
| 19 (b)                      | 3     | 9.2.2 Col 2 dot 6                                  | H6, H13           |
| 20                          | 2     | 9.3.3 Col 3 dot 4                                  | H7, H13           |
| 21 (a)                      | 3     | 9.3.1 Col 3 dot 2                                  | H9, H12, H14      |
| 21 (b)                      | 3     | 9.3.2 Col 2 dot 5                                  | H7, H9, H14       |
| 22                          | 3     | 9.3.3 Col 2 dot 3                                  | H9                |
| 23 (a)                      | 3     | 9.4.3 Col 2 dot 8                                  | H3, H1            |
| 23 (b)                      | 3     | 9.4.4 Col 2 dot 7 / Col 3 dot 5                    | H3                |
| 24                          | 6     | 9.3.3 Col 1 / Col 2 dot 3, 4, 5, Col 3 dot 2, 3    | H1, H9, H13       |
| 25                          | 6     | 9.4.2 Col 3 dot 3, 9.4.3 Col 2 dot 3               | H7, H10           |
| 26 (a)                      | 3     | 9.1. H13.1 (f)                                     | H13, H14          |

| Question                                   | Marks | Content                                                                       | Syllabus Outcomes |
|--------------------------------------------|-------|-------------------------------------------------------------------------------|-------------------|
| 26 (b)                                     | 1     | 9.1 H14.1 (f)                                                                 | H14               |
| 26 (c)                                     | 2     | 9.3.1 Col 2 dot 2 / Col 3 dot 3                                               | H14, H9           |
| 26 (d)                                     | 1     | 9.1 H14.1 (f)                                                                 | H14               |
| 27                                         | 4     | 9.1 H12 (f), 9.2.2 Col 3 dot 1                                                | H12, H9           |
| <b>Section II</b>                          |       |                                                                               |                   |
| <b>Question 28 — Geophysics</b>            |       |                                                                               |                   |
| 28 (a) (i)                                 | 1     | 9.5.1 Col 2 dot, 9.5.4 Col 2 dot 2                                            | H10               |
| 28 (a) (ii)                                | 3     | 9.5.1 Col 2 dot 1                                                             | H9                |
| 28 (b) (i)                                 | 4     | 9.5.2 Col 3 dot 3                                                             | H2, H12           |
| 28 (b) (ii)                                | 2     | 9.5.2 Col 3 dot 4, 8                                                          | H2                |
| 28 (c)                                     | 7     | 9.2.2 Col 2 dot 9, 9.5.2 Col 2 dot 3, 5, 9                                    | H4                |
| 28 (d) (i)                                 | 3     | 9.5.1 Col 2 dot 2                                                             | H12, H14          |
| 28 (d) (ii)                                | 2     | 9.5.1 Col 2 dot 1, Col 3 dot 2                                                | H14               |
| 28 (d) (iii)                               | 3     | 9.5.3 Col 2 dot 1, 2, 3, 5                                                    | H8                |
| <b>Section II</b>                          |       |                                                                               |                   |
| <b>Question 29 — Medical Physics</b>       |       |                                                                               |                   |
| 29 (a) (i)                                 | 1     | 9.6.1 Col 2 dot 2                                                             | H7, H8            |
| 29 (a) (ii)                                | 3     | 9.6.1 Col 2 dot 4, 6                                                          | H7, H8            |
| 29 (b) (i)                                 | 3     | 9.6.3 Col 2, dot 2, 3, 9.6.4 Col 3 dot 4, 5                                   | H1, H13           |
| 29 (b) (ii)                                | 3     | 9.6.1 Col 2, dot 8, 9                                                         | H4                |
| 29 (c)                                     | 7     | 9.6.4 Col 2 dot 3, 4, 5, 6, 9.4.4 Col 2 dot 7, Col 3 dot 5, 9.4.3 Col 3 dot 3 | H3, H7, H9, H10   |
| 29 (d) (i)                                 | 2     | 9.6.2 Col 2 dot 5, Col 3 dot 3                                                | H7, H8, H13       |
| 29 (d) (ii)                                | 3     | 9.6.2 Col 2 dot 6, 7                                                          | H7, H8            |
| 29 (d) (iii)                               | 3     | 9.6.2 Col 2 dot 7                                                             | H7, H8            |
| <b>Section II</b>                          |       |                                                                               |                   |
| <b>Question 30 — Astrophysics</b>          |       |                                                                               |                   |
| 30 (a) (i)                                 | 2     | 9.7.6 Col 2 dot 4                                                             | H10, H7           |
| 30 (a) (ii)                                | 2     | 9.7.6 Col 2 dot 2                                                             | H7                |
| 30 (b) (i)                                 | 3     | 9.7.4 Col 2 dot 4                                                             | H2, H14           |
| 30 (b) (ii)                                | 3     | 9.7.4 Col 3 dot 1                                                             | H2, H14           |
| 30 (c)                                     | 7     | 9.4.2 Col 2 dot 3, 9.7.3 Col 2, dot 1, 5 / Col 3 dot 1                        | H2, H7, H10       |
| 30 (d) (i)                                 | 2     | 9.7.5 Col 3 dot 1                                                             | H14               |
| 30 (d) (ii)                                | 3     | 9.7.5 Col 3 dot 2                                                             | H6                |
| 30 (d) (iii)                               | 3     | 9.7.5 Col 3 dot 1, 2                                                          | H14               |
| <b>Section II</b>                          |       |                                                                               |                   |
| <b>Question 31 — From Quanta to Quarks</b> |       |                                                                               |                   |
| 31 (a) (i)                                 | 2     | 9.8.3 Col 2 dot 8                                                             | H10               |

| Question                                | Marks | Content                                                                   | Syllabus Outcomes |
|-----------------------------------------|-------|---------------------------------------------------------------------------|-------------------|
| 31 (a) (ii)                             | 2     | 9.8.3 Col 2 dot 2                                                         | H6, H7            |
| 31 (b) (i)                              | 2     | 9.8.4 Col 2 dot 5                                                         | H10               |
| 31 (b) (ii)                             | 4     | 9.8.3 col 3 dot 2, Col 2 dot 9                                            | H7, H10           |
| 31 (c)                                  | 7     | 9.8.4 Col 2 dot 4, 5, 9.4.1 Col 2 dot 5                                   | H1, H2, H3, H6    |
| 31 (d) (i)                              | 2     | 9.8.1 Col 3 dot 1                                                         | H11               |
| 31 (d) (ii)                             | 4     | 9.8.1 Col 2 dot 5, Col 3 dot 3                                            | H7, H12           |
| 31 (d) (iii)                            | 2     | 9.8.1 Col 2 dot 6                                                         | H10               |
| <b>Section II</b>                       |       |                                                                           |                   |
| <b>Question 32 — The Age of Silicon</b> |       |                                                                           |                   |
| 32 (a) (i)                              | 2     | 9.9.4 Col 3 dot 2                                                         | H9                |
| 32 (a) (ii)                             | 2     | 9.9.4 Col 3 dot 1                                                         | H9                |
| 32 (b) (i)                              | 2     | 9.9.5 Col 2 dot 1 / Col 3 dot 2                                           | H2, H13           |
| 32 (b) (ii)                             | 4     | 9.9.5 Col 2 dot 1 / Col 3 dot 2                                           | H2, H13           |
| 32 (c)                                  | 7     | 9.9.3 Col 3 dot 8 / Col 3 dot 3, 9.9.1 Col 2 dot 1, 2, 3, 4 / Col 3 dot 1 | H1, H3, H7, H9    |
| 32 (d) (i)                              | 2     | 9.9.6 Col 2 dot 8                                                         | H7, H9, H14       |
| 32 (d) (ii)                             | 3     | 9.9.6 Col 2 dot 8 / Col 3 dot 3, 4                                        | H7, H9, H14       |
| 32 (d) (iii)                            | 3     | 9.9.6 Col 2 dot 11, 12                                                    | H9                |

## 2004 HSC Physics Marking Guidelines

### Section I, Part B

#### Question 16

*Outcomes assessed: H6, H12*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct solution given (working shown)</li> <li>Correct substitutions (both data and signs) into relevant formulae (arithmetic errors ignored)</li> </ul>                                                                                                                                                                                                                                                                                                                                            | 4     |
| <ul style="list-style-type: none"> <li>Relevant formulae used but <math>u_x</math> and <math>u_y</math> transposed consistently</li> </ul> OR <ul style="list-style-type: none"> <li>All correct but time of flight value is halved</li> </ul> OR <ul style="list-style-type: none"> <li>Lack of use of sign convention</li> </ul>                                                                                                                                                                                                          | 3     |
| <ul style="list-style-type: none"> <li>Relevant formulae used but <math>u_x</math> and <math>u_y</math> transposed inconsistently</li> </ul> OR <ul style="list-style-type: none"> <li>Angle not taken into account but +ve and -ve convention used to determine answer</li> </ul> OR <ul style="list-style-type: none"> <li>Initial velocity resolved into components correctly</li> </ul> AND <ul style="list-style-type: none"> <li>Working data has value for 'g' and initial velocity in different directions (+ve and -ve)</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Initial velocity resolved into components correctly</li> </ul> OR <ul style="list-style-type: none"> <li>Working data has value for 'g' and initial velocity in different directions (+ve and -ve)</li> </ul>                                                                                                                                                                                                                                                                                        | 1     |

### Question 17 (a) (i)

Outcomes assessed: H9, H6

#### MARKING GUIDELINES

| Criteria                                                | Marks |
|---------------------------------------------------------|-------|
| • Correctly calculates mass of the moon                 | 2     |
| • Uses correct formula but makes incorrect substitution | 1     |

### Question 17 (a) (ii)

Outcomes assessed: H9, H6

#### MARKING GUIDELINES

| Criteria                                                                           | Marks |
|------------------------------------------------------------------------------------|-------|
| • Correctly calculates the magnitude of the orbital velocity of the command module | 2     |
| • Uses correct relationship but makes incorrect substitution                       | 1     |

### Question 17 (b)

Outcomes assessed: H9, H6

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                            | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| • Any valid reasoning that leads to the independence of the mass of an object and its orbital speed and altitude                                                                    | 2     |
| • State period independent of mass or period only depends on radius/altitude<br>OR<br>• Orbital velocity depends on mass of central body not mass of orbiting body (statement only) | 1     |

### Question 18

*Outcomes assessed: H6, H14, H13*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                        | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Makes a correct judgement supported by arguments addressing horizontal forces and backed up by numerical data</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Makes a judgement insufficiently supported by argument and/or numerical data</li> </ul>                                  | 2–3   |
| <ul style="list-style-type: none"> <li>Makes correct statement about car moving in circular motion</li> </ul>                                                   | 1     |

### Question 19 (a)

*Outcomes assessed: H14, H6, H13*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Appropriately locates the positions of Earth and Mars and shows the trajectory correctly</li> </ul>                                                                                                                                                                                                   | 3     |
| <ul style="list-style-type: none"> <li>Locates Earth and Mars appropriately</li> </ul> OR <ul style="list-style-type: none"> <li>Locates Earth correctly <u>and</u> has a trajectory that forms part of a plausible ellipse (with the sun at a focus) passing through launch location of Earth and the incorrect location of Mars</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Locates Earth correctly</li> </ul> OR <ul style="list-style-type: none"> <li>Shows a plausible trajectory between incorrectly located planets i.e. orbit is plausibly elliptical around the sun</li> </ul>                                                                                            | 1     |

### Question 19 (b)

*Outcomes assessed: H6, H13*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies that velocity imparted by rocket motors to satellite is augmented by Earth's motion and provides any two of the following supporting points: <ul style="list-style-type: none"> <li>Axial rotation of Earth augmenting launch towards east</li> <li>Orbital motion of Earth (revolution) augmenting departure from Earth orbit to interplanetary trajectory</li> <li>The launch window from this position on the Earth's orbit provides the shortest trajectory</li> </ul> </li> </ul>   | 3     |
| <ul style="list-style-type: none"> <li>Identifies that the velocity imparted by rocket motors to satellite is augmented by Earth's motion and provides one of the following supporting points: <ul style="list-style-type: none"> <li>Axial rotation of Earth augmenting launch towards east</li> <li>Orbital motion of Earth (revolution) augmenting departure from Earth's orbit to interplanetary trajectory</li> <li>The launch window from this position on the Earth's orbit provides the shortest trajectory</li> </ul> </li> </ul> | 2     |
| <p>Provides one of the following supporting points:</p> <ul style="list-style-type: none"> <li>Axial rotation of Earth augmenting launch towards east</li> <li>Orbital motion of Earth (revolution) augmenting departure from Earth's orbit to interplanetary trajectory</li> <li>The launch window from this position on the Earth's orbit provides the shortest trajectory</li> </ul>                                                                                                                                                    | 1     |

## Question 20

Outcomes assessed: H7, H13

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                     | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly clarifies the role of <u>both</u> structures</li> </ul> A – protects the transmission line from lightning strikes<br>B – insulates the wires from the supporting structures | 2     |
| <ul style="list-style-type: none"> <li>Only makes clear the role of one of the structures A or B</li> </ul>                                                                                                                  | 1     |

## Question 21 (a)

Outcomes assessed: H9, H12, H14

### MARKING GUIDELINES

| Criteria                                                                                                               | Marks |
|------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly identifies THREE mistakes in the construction of the motor</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Correctly identifies TWO mistakes in the construction of the motor</li> </ul>   | 2     |
| <ul style="list-style-type: none"> <li>Correctly identifies ONE mistake in the construction of the motor</li> </ul>    | 1     |

## Question 21 (b)

Outcomes assessed: H7, H9, H14

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                        | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies link between back emf and a motor operating normally (rotation)</li> <li>Demonstrates understanding that back emf opposes supply emf</li> <li>Concluding statement that when the coil stops rotating the net emf increases and so does the current</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Link between back emf and lower current made</li> </ul>                                                                                                                                                                                                                  | 2     |
| <ul style="list-style-type: none"> <li>Back emf identified</li> </ul>                                                                                                                                                                                                                                           | 1     |

## Question 22

Outcomes assessed: H9

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies the rotor and the stator, (their relative positions) and describes one</li> <li>States that induction is responsible for the movement of the rotor (or no current passes directly to rotor or similar statement)</li> <li>States one other relevant feature (may be an expanded description relating to rotor and stator)</li> </ul>                                                              | 3     |
| <ul style="list-style-type: none"> <li>Refers to terms rotor and stator</li> </ul> OR <ul style="list-style-type: none"> <li>Describes the rotor or stator</li> </ul> OR <ul style="list-style-type: none"> <li>States that induction is responsible for the movement of the rotor (or no current passes directly to rotor or similar statement)</li> </ul> AND <ul style="list-style-type: none"> <li>States one other relevant feature</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>At least one feature</li> </ul> OR <ul style="list-style-type: none"> <li>Names rotor and stator</li> </ul>                                                                                                                                                                                                                                                                                                  | 1     |

## Question 23 (a)

Outcomes assessed: H3, H1

### MARKING GUIDELINES

| Criteria                                                                            | Marks |
|-------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Lists THREE correct disadvantages</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Lists TWO correct disadvantages</li> </ul>   | 2     |
| <ul style="list-style-type: none"> <li>Lists ONE correct disadvantage</li> </ul>    | 1     |

## Question 23 (b)

Outcomes assessed: H3

### MARKING GUIDELINES

| Criteria                                                                                                                                  | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Gives main features of ONE advantage with its respective applications</li> </ul>                   | 3     |
| <ul style="list-style-type: none"> <li>ONE advantage and ONE relevant application</li> </ul>                                              | 2     |
| <ul style="list-style-type: none"> <li>ONE advantage only</li> </ul> OR <ul style="list-style-type: none"> <li>ONE application</li> </ul> | 1     |

## Question 24

*Outcomes assessed: H1, H9, H13*

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                             | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Well presented arguments discussing the advantages of Westinghouse's AC system of generation and distribution</li> <li>Refers to both generation and distribution over longer distances</li> </ul>                                                                                            | 5–6   |
| <ul style="list-style-type: none"> <li>Outlines advantages of AC over DC</li> <li>Makes references to generation and distribution</li> <li>Provides some reasons to support argument</li> </ul>                                                                                                                                      | 4     |
| <ul style="list-style-type: none"> <li>Outlines advantages of AC and DC</li> <li>Makes well presented arguments for generation or distribution</li> <li>Provides some reasons to support argument</li> </ul>                                                                                                                         | 3     |
| <ul style="list-style-type: none"> <li>Answer gives attributes of AC but no reference to advantages over DC</li> <li>Supporting statements are lists only</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>Features of AC discussed with brief outline but no clear evidence of understanding why an advantage</li> </ul> | 1–2   |

## Question 25

Outcomes assessed: H7, H10

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                        | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies the photo-electric effect on semiconductors, the generation of free electrons and holes, their diffusion, the creation of a potential difference, and the production of a current</li> </ul>                                  | 6     |
| <ul style="list-style-type: none"> <li>Identifies the photo-electric effect, the generation of free electrons and holes, the creation of a potential difference or electron diffusion and the production of a current</li> </ul>                                                | 5     |
| <ul style="list-style-type: none"> <li>Identifies the photo-electric effect, the generation of free electrons and holes, and of current flowing through the circuit</li> </ul>                                                                                                  | 4     |
| <ul style="list-style-type: none"> <li>Identifies the photo-electric effect and current flowing</li> </ul> OR <ul style="list-style-type: none"> <li>Outlines properties of <math>p</math> and <math>n</math> type layers with reference to free electrons and holes</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Identifies that the current flows during the generation of free electrons/holes</li> </ul> OR <ul style="list-style-type: none"> <li>Outlines properties of <math>p</math> and <math>n</math> type layers</li> </ul>                     | 2     |
| <ul style="list-style-type: none"> <li>Identify that free electrons are generated</li> </ul> OR <ul style="list-style-type: none"> <li>A simple energy transformation</li> </ul> OR <ul style="list-style-type: none"> <li>Identifies the photo-electric effect</li> </ul>      | 1     |

## Question 26 (a)

Outcomes assessed: H13, H14

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                    | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly plots the 5 points</li> <li>Correctly draws line of best fit that indicates an understanding of what this is</li> </ul>                                                                                                    | 3     |
| <ul style="list-style-type: none"> <li>Correctly plots the 5 points</li> <li>Fails to draw line of best fit</li> </ul> OR <ul style="list-style-type: none"> <li>Correctly draws a line of best fit through the plotted points of which at least two are correct</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>At least two are correctly plotted and no line of best fit or wrong line</li> </ul>                                                                                                                                                  | 1     |

### Question 26 (b)

Outcomes assessed: H14

#### MARKING GUIDELINES

| Criteria                                                                                                  | Marks |
|-----------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Gradient correctly determined from the line of best fit</li> </ul> | 1     |

### Question 26 (c)

Outcomes assessed: H14, H9

#### MARKING GUIDELINES

| Criteria                                                                                                                                   | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Answer demonstrates an understanding of relationship between <math>k</math> and gradient</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Correctly identifies the formula required</li> </ul>                                                | 1     |

### Question 26 (d)

Outcomes assessed: H14

#### MARKING GUIDELINES

| Criteria                                                                                                                                                           | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly determines <math>k</math> using the (value of the) gradient from part (b) and the equation in part (c)</li> </ul> | 1     |

## Question 27

Outcomes assessed: H12, H9

### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly determines the take off speed and recognises that this is impossible OR correctly determines that the height to which the athlete jumps is impossible AND hence the information is not accurate</li> </ul>                                                                                                                                                                                                                                       | 4     |
| <ul style="list-style-type: none"> <li>Correctly determines the take off speed and recognises that this is impossible OR correctly determines that the height to which the athlete jumps is impossible BUT does NOT make an assessment of the article</li> </ul>                                                                                                                                                                                                                                  | 3     |
| <ul style="list-style-type: none"> <li>Makes a correct calculation but does not recognise the answer as being impossible nor the flawed nature of the article</li> </ul> OR <ul style="list-style-type: none"> <li>Makes a conclusion based on incorrect value for the time of flight</li> </ul> OR <ul style="list-style-type: none"> <li>Makes an incorrect substitution into a correct equation with conclusion and assessment of the article consistent with the calculated values</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>States that staying in the air for 2.5 seconds is impossible, with no justification</li> </ul> OR <ul style="list-style-type: none"> <li>Makes a correct statement</li> </ul> OR <ul style="list-style-type: none"> <li>Substitutes incorrect time of flight</li> </ul>                                                                                                                                                                                    | 1     |

## Section II

### Question 28 (a) (i)

*Outcomes assessed: H10*

#### MARKING GUIDELINES

| Criteria                                                                                   | Marks |
|--------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>TWO properties of earth materials listed</li> </ul> | 1     |

### Question 28 (a) (ii)

*Outcomes assessed: H9*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                          | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Description of alignment of magnetic minerals in rocks related to Earth's magnetic field at times of formation</li> <li>Observation of reversals indicates magnetic field reversal occurring on a regular basis</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>As above, but talk about rocks rather than magnetic minerals</li> </ul> OR <ul style="list-style-type: none"> <li>Describes alignment reversals but does not link to magnetic field (or vice versa)</li> </ul>             | 2     |
| <ul style="list-style-type: none"> <li>Mention that Earth's magnetic field has been reversed in the past</li> </ul> OR <ul style="list-style-type: none"> <li>Magnetic minerals in rocks align with Earth's magnetic field</li> </ul>                             | 1     |

### Question 28 (b) (i)

Outcomes assessed: H2, H12

#### MARKING GUIDELINES

| Criteria                                                                                                                                           | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| • Correct value for radius determined                                                                                                              | 4     |
| • Correct value of 'g' obtained through correct substitution<br>• Attempt at finding 'r' made using calculated value of 'g'                        | 3     |
| • Correct value of 'g' obtained through correct substitution<br>OR<br>• Data sheet value of 'g' used and correct substitution into determining 'r' | 2     |
| • Attempt to find 'g'<br>OR<br>• Attempt made at finding 'r' using 'g' from data sheet                                                             | 1     |

### Question 28 (b) (ii)

Outcomes assessed: H2

#### MARKING GUIDELINES

| Criteria                                                                                                                                                        | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| • Links longer period to smaller value of 'g' (may be inferred)<br>• Provides a correct reasoning for variation consistent with strength of gravitational field | 2     |
| • Indicates that 'g' is smaller<br>OR<br>• Links a value of 'g' with strength of gravitational field                                                            | 1     |

### Question 28 (c)

Outcomes assessed: H4

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                        | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Describes how satellites provide information about the Earth by remote sensing using geostationary and low Earth orbit satellites. Puts forward valid reasons for the preference of one type of satellite for the collection of specific data</li> </ul> | 6–7   |
| <ul style="list-style-type: none"> <li>Describes the type of information provided by satellites</li> <li>Makes some comparisons of geostationary and low Earth orbit satellites</li> <li>Outlines reasons for preferring one type of satellite over the other</li> </ul>                        | 5     |
| <ul style="list-style-type: none"> <li>Outlines the type of information provided by satellites</li> <li>Makes some comparisons of geostationary and low Earth orbit satellites</li> <li>Outlines reasons for preferring one type of satellite over the other</li> </ul>                         | 4     |
| <ul style="list-style-type: none"> <li>Outlines the type of information provided by satellites</li> <li>Outlines some features of geostationary and/or low Earth orbit satellites</li> </ul>                                                                                                    | 3     |
| <ul style="list-style-type: none"> <li>Identifies the type of information provided by satellites</li> <li>Identifies some features of geostationary and/or low Earth orbit satellites</li> </ul>                                                                                                | 2     |
| <ul style="list-style-type: none"> <li>Identifies a correct fact about satellites</li> </ul> OR <ul style="list-style-type: none"> <li>Identifies at least two types of information that can be obtained from remote sensing</li> </ul>                                                         | 1     |

### Question 28 (d) (i)

Outcomes assessed: H12, H14

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                       | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Outline of experimental method used which shows familiarity with doing experiments</li> <li>Evidence of reliability from experimental method</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Outline of experimental method used which shows familiarity with experiments but no clear links to reliability</li> </ul>                               | 2     |
| <ul style="list-style-type: none"> <li>Outline of experimental method only</li> </ul>                                                                                                          | 1     |

### Question 28 (d) (ii)

Outcomes assessed: H14

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                  | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Discontinuity indicates changes in composition of rock type and or effects of temperature and pressure on structures</li> </ul> OR | 2     |
| <ul style="list-style-type: none"> <li>Relate above answer to specific layer of Earth's internal structure</li> </ul>                                                     |       |
| <ul style="list-style-type: none"> <li>Recognition of reason for discontinuity at either of named points</li> </ul>                                                       | 1     |

### Question 28 (d) (iii)

Outcomes assessed: H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Link velocity changes in P &amp; S waves to variation in density as shown on graph</li> <li>Identify 50km as site of change in structure / density therefore refraction occurring (may relate to 2895 instead)</li> <li>Identify 2895km as change of state to liquid due to loss in velocity of P waves and stopping of S waves</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>TWO of above points</li> </ul>                                                                                                                                                                                                                                                                                                             | 2     |
| <ul style="list-style-type: none"> <li>ONE of above points</li> </ul>                                                                                                                                                                                                                                                                                                             | 1     |

### Question 29 (a) (i)

Outcomes assessed: H7, H8

#### MARKING GUIDELINES

| Criteria                                                                                                                         | Marks |
|----------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Alternating potential difference (or voltage) applied to piezoelectric crystal</li> </ul> | 1     |

### Question 29 (a) (ii)

Outcomes assessed: H7, H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                            | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Reflection of ultrasound at tissue boundary</li> <li>• Distances calculated from time for signal to return to ultrasound transducer</li> <li>• The brightness difference is greater if the difference between acoustic impedance is greater such as, flesh / bone showing up the skulls clearly</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>• Reflection of ultrasound at skull boundary and mentions the differences in acoustic impedance or an indication that this is an ultrasound</li> </ul>                                                                                                                                                       | 2     |
| <ul style="list-style-type: none"> <li>• Sound waves reflect off the heads/skulls OR indicates that this is an ultrasound</li> </ul>                                                                                                                                                                                                                | 1     |

### Question 29 (b) (i)

Outcomes assessed: H1, H13

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Makes a clear judgement based on the functional nature of PET providing information about the location and level of activity and location of areas (structure and function) in the brain associated with tasks such as seeing and learning or other brain functions</li> </ul> <p>AND</p> <ul style="list-style-type: none"> <li>• States that glucose concentrates more in areas of greater metabolic activity</li> <li>• Tagged glucose produces positrons which interact with electrons to produce gamma rays</li> <li>• These gamma rays can be used to indicate areas of activity</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>• States that glucose concentrates more in areas of greater metabolic activity</li> <li>• Tagged glucose produces positrons which interact with electrons to produce gamma rays</li> <li>• These gamma rays can be used to indicate areas of activity</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• Makes a judgement and states any two of the points above</li> </ul>                                                                                                                                                                                                             | 2     |
| <ul style="list-style-type: none"> <li>• PET is functional</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• F-18 is a gamma-source or positron source</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• PET shows active areas</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• Glucose is used in greater amounts by more active areas</li> </ul>                                                                                                                                                                                                                                                   | 1     |

### Question 29 (b) (ii)

*Outcomes assessed: H4*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies the process as Doppler ultrasound</li> </ul> AND <ul style="list-style-type: none"> <li>Describes the movement of blood towards transducer (or Ultrasound head or observer) as increasing the frequency and movement away decreases frequency</li> </ul> AND <ul style="list-style-type: none"> <li>States that the magnitude (or amount) of the change depends on the relative speed</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Identifies the process as Doppler ultrasound and states that movement of blood produces a frequency change of the ultrasound</li> </ul>                                                                                                                                                                                                                                                                     | 2     |
| <ul style="list-style-type: none"> <li>Identifies the process as being Doppler ultrasound</li> </ul> OR <ul style="list-style-type: none"> <li>Description of the Doppler effect</li> </ul>                                                                                                                                                                                                                                                        | 1     |

### Question 29 (c)

*Outcomes assessed: H3, H7, H9, H10*

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Makes a judgement supported by evidence linked to identified advances in knowledge about both semiconductors and superconductors which substantiates their impact on MRI development</li> <li>Describes the physical basis of MRI</li> </ul>                                                                                                                                                                   | 6–7   |
| <ul style="list-style-type: none"> <li>Describes the physical basis of MRI and</li> <li>Makes a judgment about the impact/advances of semiconductors and superconductors but does not support the assessment with clear, specific facts</li> </ul> OR <ul style="list-style-type: none"> <li>Makes a judgement about the impact/advances of semiconductors or superconductors with evidence substantiating their impact on MRI development</li> </ul> | 5     |
| <ul style="list-style-type: none"> <li>Describes the physical basis of MRI</li> <li>Makes a judgement about the impact/advances of semiconductors and superconductors with limited support for one technology</li> </ul>                                                                                                                                                                                                                              | 4     |
| <ul style="list-style-type: none"> <li>Outline the physical basis of the MRI</li> </ul> OR <ul style="list-style-type: none"> <li>Outlines parts of the MRI process with parts omitted</li> </ul> AND <ul style="list-style-type: none"> <li>Identifies advances in semiconductor or superconductor technology</li> </ul>                                                                                                                             | 3     |
| <ul style="list-style-type: none"> <li>Outlines parts of the MRI process with parts omitted</li> </ul> OR <ul style="list-style-type: none"> <li>Identifies advances in semiconductor or superconductor technology</li> </ul>                                                                                                                                                                                                                         | 2     |
| <ul style="list-style-type: none"> <li>Identifies a correct fact about MRI or semiconductor or superconductor technology</li> </ul>                                                                                                                                                                                                                                                                                                                   | 1     |

### Question 29 (d) (i)

Outcomes assessed: H7, H8, H13

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                      | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Ray of light strikes P at an angle greater than <math>I_c</math> as measured from the normal</li> </ul> AND <ul style="list-style-type: none"> <li>Angle of reflection = angle of incidence</li> </ul> | 2     |
| Any <ul style="list-style-type: none"> <li>Reflected ray with angle of incidence = angle of reflection</li> </ul>                                                                                                                             | 1     |

### Question 29 (d) (ii)

Outcomes assessed: H7, H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Describes the transmission of light to the area being imaged by non-coherent fibre bundles and the return of the image via coherent fibres including a description of the role of coherent fibres in maintaining the integrity of the image</li> </ul>                                                                                                                                                                   | 3     |
| Any two of the following points: <ul style="list-style-type: none"> <li>Describes the transmission of light to the area being imaged by non-coherent fibre bundles</li> <li>Describes the return of the image via coherent fibre bundles</li> <li>Describes the role of the coherent fibres in maintaining the integrity of the image</li> </ul>                                                                                                                | 2     |
| <ul style="list-style-type: none"> <li>Describes the role of the total internal reflection in the transmission of light to the optical fibres</li> </ul> OR <ul style="list-style-type: none"> <li>Describes the light returning from the object being viewed via coherent bundle of fibre</li> </ul> OR <ul style="list-style-type: none"> <li>Describes the overall process of transmission, reflection and return of the light creating the image</li> </ul> | 1     |

### Question 29 (d) (iii)

Outcomes assessed: H7, H8

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                 | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Describes the incorporation of tissue sampling instruments into the endoscope</li> </ul> AND <ul style="list-style-type: none"> <li>TWO reasons for use (eg real time, real colour, minimally invasive cost effective)</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Describes the incorporation of tissue sampling instruments into the endoscope</li> </ul> AND <ul style="list-style-type: none"> <li>ONE reason for use</li> </ul>                                                                 | 2     |
| <ul style="list-style-type: none"> <li>ONE reason for use</li> </ul> OR <ul style="list-style-type: none"> <li>One description of use</li> </ul>                                                                                                                         | 1     |

### Question 30 (a) (i)

Outcomes assessed: H10, H7

#### MARKING GUIDELINES

| Criteria                                                                             | Marks |
|--------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies both elements correctly</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Identifies ONE element correctly</li> </ul>   | 1     |

### Question 30 (a) (ii)

Outcomes assessed: H7

#### MARKING GUIDELINES

| Criteria                                                                                                            | Marks |
|---------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies type of star and mode of nuclear burning</li> </ul>               | 2     |
| <ul style="list-style-type: none"> <li>Identifies either the type of star or the mode of nuclear burning</li> </ul> | 1     |

### Question 30 (b) (i)

Outcomes assessed: H2, H14

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                                                                                                            | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Calculates correctly the colour index of each of the THREE stars</li> <li>Identify that the bluest star has the smallest value of the colour index</li> </ul>                                                                                                                                | 3     |
| <ul style="list-style-type: none"> <li>Calculates correctly the colour index of least one star and</li> <li>Identifies that bluest star has smallest colour index</li> </ul> OR <ul style="list-style-type: none"> <li>Calculates correctly the colour index of each of the THREE stars but identifies the wrong star</li> </ul>    | 2     |
| <ul style="list-style-type: none"> <li>Shows understanding of the concept of colour index, but incorrectly determines its value</li> </ul> OR <ul style="list-style-type: none"> <li>Correctly calculates one value of the colour index</li> </ul> OR <ul style="list-style-type: none"> <li>Identifies the correct star</li> </ul> | 1     |

### Question 30 (b) (ii)

Outcomes assessed: H2, H14

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                              | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies correct formula to apply, rearranges equation correctly, substitutes the appropriate values from the table</li> </ul>                                                               | 3     |
| <ul style="list-style-type: none"> <li>Identifies correct formula, but either rearrange equation incorrectly, substitutes in a wrong value</li> </ul> OR <ul style="list-style-type: none"> <li>Makes a substitution error</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Identifies correct formula, but makes two mistakes in rearranging or substituting</li> </ul>                                                                                                   | 1     |

### Question 30 (c)

Outcomes assessed: H2, H7, H10

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                      | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly describes at least one feature of the spectrum of a star, and links these to determine the temperature, chemical composition, rotational and transitional velocity of the star</li> </ul>    | 6–7   |
| <ul style="list-style-type: none"> <li>Describes or outlines at least one feature of the spectrum, and correctly describes how it/they may be used to determine 2 characteristics or outlines all three characteristics</li> </ul>            | 4–5   |
| <ul style="list-style-type: none"> <li>Outlines at least one feature and correctly outlines how to determine two of the characteristics</li> </ul>                                                                                            | 3     |
| <ul style="list-style-type: none"> <li>Outlines a feature of a star and outlines how one characteristic is determined</li> </ul> OR <ul style="list-style-type: none"> <li>Outlines how two or more characteristics are determined</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Identifies features of the spectrum of a star</li> </ul> OR <ul style="list-style-type: none"> <li>Identifies how ONE characteristic is determined</li> </ul>                                          | 1     |

### Question 30 (d) (i)

Outcomes assessed: H14

#### MARKING GUIDELINES

| Criteria                                                                                                                        | Marks |
|---------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Identifies TWO features which provide evidence for an eclipsing binary system</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Identifies ONE of the features</li> </ul>                                                | 1     |

### Question 30 (d) (ii)

Outcomes assessed: H6

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                         | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Determines orbital period from graph</li> <li>Correct substitution into correct formula</li> </ul>                                                        | 3     |
| <ul style="list-style-type: none"> <li>Determines orbital period from graph</li> <li>Identifies correct formula or</li> <li>Makes one incorrect substitution into the correct formula</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Identifies the correct equation</li> </ul> OR <ul style="list-style-type: none"> <li>Determines orbital period from graph</li> </ul>                      | 1     |

### Question 30 (d) (iii)

Outcomes assessed: H14

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                         | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Relates amount of dip in light curve to fraction blocking light from star</li> <li>Calculates area blocked by the star (or intensity ratios) to determine the fractional ratio</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Identifies that dip in light curve is due to blocking of star; but incorrectly calculates the relationships</li> </ul>                                                                    | 2     |
| <ul style="list-style-type: none"> <li>Identifies that dip in light curve is related to blocking of light</li> </ul> OR <ul style="list-style-type: none"> <li>Identifies a relevant mathematical relationship</li> </ul>        | 1     |

### Question 31 (a) (i)

Outcomes assessed: H10

#### MARKING GUIDELINES

| Criteria                                                                            | Marks |
|-------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Two features correctly identified</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>One feature correctly identified</li> </ul>  | 1     |

### Question 31 (a) (ii)

Outcomes assessed: H6, H7

#### MARKING GUIDELINES

| Criteria             | Marks |
|----------------------|-------|
| • Names the two laws | 2     |
| • Names one law      | 1     |

### Question 31 (b) (i)

Outcomes assessed: H10

#### MARKING GUIDELINES

| Criteria                                                                      | Marks |
|-------------------------------------------------------------------------------|-------|
| • Correctly states the quark composition of the neutron AND the negative pion | 2     |
| • Correctly states the composition of either                                  | 1     |

### Question 31 (b) (ii)

Outcomes assessed: H7, H10

#### MARKING GUIDELINES

| Criteria                                                                          | Marks |
|-----------------------------------------------------------------------------------|-------|
| • Correctly substitutes data in the correct equation to determine energy released | 4     |
| • Identifies correct energy equation AND<br>• Mass defect                         | 3     |
| • Calculate the mass defect                                                       | 2     |
| • Calculate the sums of the masses of reactants OR products                       | 1     |

### Question 31 (c)

Outcomes assessed: H1, H2, H3, H6

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                             | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Description of the key features and components of the standard model of matter and</li> <li>Links these to the ways accelerators are used as probes</li> </ul>                | 6–7   |
| <ul style="list-style-type: none"> <li>Describes key features and components of the standard model of matter</li> <li>Outlines how accelerators are used as probes</li> </ul>                                        | 4–5   |
| <ul style="list-style-type: none"> <li>Outlines the key features or components of the standard model of matter</li> <li>Outlines way accelerators have been used as probes</li> </ul>                                | 2–3   |
| <ul style="list-style-type: none"> <li>Identifies a correct feature of the standard model of matter</li> </ul> OR <ul style="list-style-type: none"> <li>Outlines the way accelerators are used as probes</li> </ul> | 1     |

### Question 31 (d) (i)

Outcomes assessed: H11

#### MARKING GUIDELINES

| Criteria                                                                                                | Marks |
|---------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly identifies the two pieces of equipment used</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Correctly identifies one piece of equipment used</li> </ul>      | 1     |

### Question 31 (d) (ii)

Outcomes assessed: H7, H12

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                       | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly identify appropriate equations and correct substitution</li> </ul>                                                                                            | 4     |
| <ul style="list-style-type: none"> <li>Correct substitution into Balmer/Rydberg equation</li> </ul> AND <ul style="list-style-type: none"> <li>Identification of additional equations required</li> </ul>      | 3     |
| <ul style="list-style-type: none"> <li>Identifies relevant equations</li> </ul> OR <ul style="list-style-type: none"> <li>Correct substitution into Balmer/Rydberg equation to determine wavelength</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Identifies one of the correct equations required</li> </ul>                                                                                                             | 1     |

### Question 31 (d) (iii)

Outcomes assessed: H10

#### MARKING GUIDELINES

| Criteria                    | Marks |
|-----------------------------|-------|
| • Two limitations described | 2     |
| • One limitation described  | 1     |

### Question 32 (a) (i)

Outcomes assessed: H9

#### MARKING GUIDELINES

| Criteria                                     | Marks |
|----------------------------------------------|-------|
| • Correctly outlines role of both components | 2     |
| • Correctly outlines role of ONE             | 1     |

### Question 32 (a) (ii)

Outcomes assessed: H9

#### MARKING GUIDELINES

| Criteria                 | Marks |
|--------------------------|-------|
| • Links cause and effect | 2     |
| • States cause or effect | 1     |

### Question 32 (b)

Outcomes assessed: H2, H13

#### MARKING GUIDELINES

| Criteria                                                                      | Marks |
|-------------------------------------------------------------------------------|-------|
| • Identify gate correctly<br>AND                                              | 2     |
| • Determine output correctly                                                  |       |
| • Identify gate incorrectly but match output correctly to this identification | 1     |

### Question 32 (b) (ii)

Outcomes assessed: H2, H13

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correctly determines that the gate should be an 'OR' and correctly determines the truth table</li> </ul>                                                                         | 4     |
| <ul style="list-style-type: none"> <li>Correctly determines the truth table but chooses wrong gate</li> </ul> OR <ul style="list-style-type: none"> <li>Chooses wrong gate due to one mistake in truth table</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Makes more than one error in truth table</li> </ul>                                                                                                                              | 2     |
| <ul style="list-style-type: none"> <li>Correctly states output of C</li> </ul> OR <ul style="list-style-type: none"> <li>States the correct gate</li> </ul>                                                             | 1     |

### Question 32 (c)

Outcomes assessed: H1, H3, H7, H9

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                                   | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Makes a judgement supported by a description of the properties of the three types of devices and the affect on the computing power due to a higher concentration of components on a chip</li> </ul> | 6–7   |
| <ul style="list-style-type: none"> <li>Describes the properties of the three types of devices and makes unsupported statements on the effect on computing power</li> </ul>                                                                 | 4–5   |
| <ul style="list-style-type: none"> <li>Outlines properties of the three types of devices</li> </ul>                                                                                                                                        | 3     |
| <ul style="list-style-type: none"> <li>Outlines properties of at least two devices</li> </ul> OR <ul style="list-style-type: none"> <li>Outlines developments in computers</li> </ul>                                                      | 2     |
| <ul style="list-style-type: none"> <li>Identifies one correct property of a device or a development in computers</li> </ul>                                                                                                                | 1     |

### Question 32 (d) (i)

Outcomes assessed: H7, H9, H14

#### MARKING GUIDELINES

| Criteria                                                   | Marks |
|------------------------------------------------------------|-------|
| • Correctly determines $V_{out}$                           | 2     |
| • Uses correct formula but makes a mistake in substitution | 1     |

### Question 32 (d) (ii)

Outcomes assessed: H7, H9, H14

#### MARKING GUIDELINES

| Criteria                                                                        | Marks |
|---------------------------------------------------------------------------------|-------|
| • Correctly calculates and verifies $V_{out}$                                   | 3     |
| • Correctly calculates $V_{out}$<br>OR<br>• Correctly verifies $V_{out}$        | 2     |
| • Attempts to calculate $V_{out}$ using the results from part (a) and Diagram 2 | 1     |

### Question 32 (d) (iii)

Outcomes assessed: H9

#### MARKING GUIDELINES

| Criteria                                                                                                                                                                                                                          | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| • States input resistors provide summed input current<br>• Identifies the 300k resistor as the negative feedback resistor<br>• Explains amplifier output provides exactly opposing feedback current through the feedback resistor | 3     |
| • Two of the above points                                                                                                                                                                                                         | 2     |
| • One of the above points                                                                                                                                                                                                         | 1     |